MATHEMATICS GRADE 10: AREA OF POLYGONS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.2 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 2.0: Geometry
Sub-Strand	Sub-Strand 2.5: Area of Polygons
Total Duration	8 lessons (approximately 16 periods × 40 minutes = 640 minutes)
Sub-Strand Content	– Area of a triangle using the formula ½ × base × height – Area of a triangle using the sine formula: ½ab sin(θ) – Area of a triangle using Heron's formula: A = √[s(s−a)(s−b)(s−c)] – Properties and classification of polygons (regular and irregular) – Area of regular polygons (equilateral triangle, square, pentagon, hexagon, octagon) – Area of irregular polygons (decomposition method) – Application of polygon area formulae in real-world contexts – Problem solving involving combined and composite polygon shapes
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) calculate the area of a triangle using the formula ½ × base × height in different situations, b) derive and apply the sine formula ½ab sin(θ) to calculate the area of a triangle given two sides and an included angle, c) apply Heron's formula to calculate the area of a triangle given all three sides, d) identify and classify polygons as regular or irregular based on their properties, e) calculate the area of regular polygons using appropriate formulae, f) calculate the area of irregular polygons by decomposing them into triangles and other known shapes, g) solve real-life problems involving areas of polygons in different contexts, h) appreciate the application of area of polygons in day-to-day activities such as land surveying, architecture, and design.
Core Competencies	– Communication and Collaboration: The learner discusses with peers to derive and verify area formulae for triangles and polygons, sharing strategies for decomposing irregular shapes into manageable parts. – Critical Thinking and Problem Solving: The learner selects the most appropriate area formula (½bh, sine rule, or Heron's) depending on the given information, and applies decomposition strategies to irregular polygons. – Digital Literacy: The learner uses digital devices and video resources to explore and verify area calculations for triangles and regular polygons, including interactive simulations. – Creativity and Imagination: The learner designs composite polygon shapes and calculates their areas, connecting mathematical concepts to real-world design and construction contexts in Kenya. – Learning to Learn: The learner reflects on which formula is most efficient for a given triangle scenario, building metacognitive awareness of mathematical tool selection.

Core Values	– Responsibility: The learner takes care of mathematical tools (rulers, protractors, calculators) used during area investigations and polygon constructions. – Unity: The learner works collaboratively in groups to decompose irregular polygons and verify area calculations, valuing each member's contribution. – Respect: The learner listens to and respects peers' alternative methods for calculating areas, recognising that multiple valid approaches exist. – Social Justice: The learner appreciates how accurate land area measurement is critical for fair land distribution and ownership in Kenyan communities, particularly in rural and peri-urban areas.
Science & Engineering Practices	– Developing and Using Models: The learner constructs visual and algebraic models of triangles and polygons to represent and calculate area, revising models as new formulae are introduced. – Using Mathematics and Computational Thinking: The learner applies and compares three distinct area formulae for triangles and extends computational strategies to regular and irregular polygons. – Analysing and Interpreting Data: The learner measures dimensions of physical polygon shapes, records data, calculates areas, and evaluates the accuracy of different methods. – Constructing Explanations: The learner explains why different formulae are needed depending on the information available (height, angles, or all three sides), constructing reasoned justifications. – Obtaining, Evaluating, and Communicating Information: The learner watches instructional videos, reads worked examples, and communicates findings on polygon areas to peers through presentations and model displays.
Pertinent & Contemporary Issues (PCIs)	– Life Skills (Critical Thinking): The learner evaluates which area formula is most efficient given specific triangle conditions, developing decision-making and analytical life skills. – Life Skills (Problem Solving): The learner applies area of polygons to solve practical problems such as estimating the area of a shamba (farm plot), a roof section, or a tiled floor pattern in a Kenyan household. – Citizenship: The learner connects polygon area calculations to land surveying and title deed demarcation in Kenya, understanding the civic importance of accurate measurement. – Environmental Education: The learner explores how area calculations are used in conservation planning, such as estimating the size of forest reserves or wildlife corridors in Kenya's national parks.
Career Connections	– Land Surveyor: Area of polygons is fundamental to surveying and demarcating land boundaries for farms, plots, and estates across Kenya's counties. – Architect and Civil Engineer: Calculating areas of triangular and polygonal floor plans, roof sections, and structural components is a core professional skill in building design in Kenya. – Quantity Surveyor: Estimating material quantities (tiles, roofing sheets, paint) for construction projects requires accurate polygon area calculations. – Graphic Designer and Artist: Regular and irregular polygon area knowledge informs pattern design, tiling, and visual layout work in Kenya's creative industries. – Agricultural Extension Officer: Estimating the area of irregularly shaped farm plots (shambas) helps farmers in counties like Kericho, Trans Nzoia, and Nakuru plan crop yields and input use.
Focus for Lessons	This unit develops learners' ability to calculate areas of triangles using three distinct methods — the base-height formula, the sine formula, and Heron's formula — and extends these skills to computing the areas of regular and irregular polygons. Using the anchoring phenomenon of a Kenyan land surveyor measuring an irregularly shaped farm plot near the Rift Valley, learners are challenged to discover why a single formula is not always sufficient and how different mathematical tools unlock solutions depending on available information. Throughout the unit, real-world Kenyan contexts — including architectural designs, tiled mosque floors in Mombasa, and shamba demarcation in Trans Nzoia — ground every lesson in authentic problem-solving.

Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot, no matter what measurements are available? KICD KEY INQUIRY QUESTIONS: 1. How do we determine the area of a polygon? 2. How do we apply area of polygons in real-life situations?
Anchoring Phenomenon	A land surveyor in Nakuru County has been hired to determine the exact area of an irregularly shaped shamba (farm plot) left to three siblings after their father's passing. The plot has five boundary corners marked by old fence posts, and no two adjacent sides are equal in length — the plot cannot be neatly described as a rectangle or a simple triangle. The surveyor has measurements of some sides and some angles but not all heights. Standing at the edge of the plot, the surveyor must decide: which formula do I use when I don't know the height of a triangle? Can I even calculate the area of this five-sided irregular plot? Students observe that the surveyor's standard ½ × base × height formula cannot be directly applied without a perpendicular height, and that the plot must be broken into triangles to be solved. This puzzling situation — how do you find the area of something you cannot neatly measure with a ruler alone? — anchors the entire unit and motivates the exploration of the sine formula, Heron's formula, and polygon decomposition strategies.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Area of a Triangle (½ × base × height): Students are introduced to the Nakuru land surveyor phenomenon. They predict how they would measure the shamba's area using only a tape measure. Learners review and apply the foundational ½ × base × height formula through hands-on measurement of triangular shapes on squared paper, connecting to the surveyor's challenge of needing a perpendicular height. Lesson 2 — Area of a Triangle Using the Sine Formula (½ab sinθ): Students investigate what happens when the height cannot be measured directly. Through a guided derivation, learners discover that ½ab sin(θ) gives the area when two sides and the included angle are known. Worked examples use triangular farm plots with given side lengths and angles, directly addressing the surveyor's dilemma. Lesson 3 — Applying the Sine Formula & Video Examples: Learners watch a video demonstrating the sine formula in action and work through a series of Kenyan-context examples (triangular roof sections in Eldoret, boat sail areas on Lake Victoria). They practise selecting between ½bh and ½ab sinθ based on given information. Lesson 4 — Area of a Triangle Using Heron's Formula: Students explore the scenario of the Kisumu carpenter who knows all three sides but no angle or height. Learners derive and apply Heron's formula $A = \sqrt{s(s-a)(s-b)(s-c)}$, verifying results against the sine formula where possible. Class discussion addresses which formula is most powerful and why. Lesson 5 — Polygons: Classification and Properties: Students classify polygons as regular or irregular, convex or concave, using cut-out shapes and video resources. Learners connect this to the five-sided shamba plot and realise any polygon can be decomposed into triangles — the key insight for solving the anchoring phenomenon. Lesson 6 — Area of Regular Polygons: Learners derive and apply formulae for the area of regular polygons (equilateral triangle, square, regular pentagon, regular hexagon, regular octagon) using the central triangle decomposition method. Supporting phenomenon: calculating tile coverage for the Jamia Mosque hexagonal floor pattern. Lesson 7 — Area of Irregular Polygons: Students return fully to the anchoring phenomenon and decompose the surveyor's five-

	sided shamba plot into triangles, applying all three triangle area formulae as needed. Learners solve the agricultural officer's quadrilateral farm plot problem and the Eldoret school's pentagonal mural wall, consolidating all prior learning. Lesson 8 — Consolidation, Model Revision & Real-World Applications: Learners revise their running polygon area model, present solutions to the driving question, and tackle mixed-context problems combining regular and irregular polygons. A class Driving Question Board review documents the full journey from ½bh to composite polygon decomposition, celebrating how mathematical tools unlock real-world land and design problems across Kenya.
Supporting Phenomena	– A tiled floor in the Jamia Mosque in Nairobi features a repeating pattern of regular hexagonal and triangular tiles — how do the tilers calculate exactly how many tiles of each shape are needed to cover the entire prayer hall floor without waste? – A carpenter in Kisumu is building a triangular wooden tabletop for a client and knows the lengths of all three sides but has no way to measure the height — how does she calculate how much varnish to buy to coat the surface? – A school in Eldoret wants to paint a large mural on a pentagonal wall section — the art teacher must calculate the area of the irregular pentagon to estimate how many tins of paint are needed. – A maize farmer in Trans Nzoia has a plot shaped like a quadrilateral with no right angles; the local agricultural officer needs to estimate the plot's area to advise on the correct amount of fertiliser to purchase per acre.

LESSON 1 (80 minutes): Anchoring Phenomenon & Area of a Triangle (½ × base × height)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce the anchoring phenomenon of the Nakuru land surveyor and establish the foundational ½ × base × height formula as the entry point into the unit's central challenge.
Knowledge	By the end of the lesson, the learner should be able to recall and state the formula for the area of a triangle as Area = ½ × base × height, and identify the perpendicular height as a necessary component of the formula.
Skills	By the end of the lesson, the learner should be able to measure and identify the base and perpendicular height of triangles drawn on squared paper, and calculate the area of triangles using the formula Area = ½ × base × height.
Attitudes	By the end of the lesson, the learner should be able to appreciate the relevance of triangle area calculations in real-life land measurement contexts such as farm plot surveying in Kenya.
Key Inquiry Question	How do we determine the area of a polygon?
Purpose in Storyline	This opening lesson launches the unit phenomenon: a land surveyor in Nakuru County must determine the exact area of an irregularly shaped five-sided shamba inherited by three siblings. Students are introduced to the surveyor's dilemma — the plot cannot be measured as a simple rectangle — and make their first predictions about how area could be calculated. They then revisit the foundational ½ × base × height formula through hands-on measurement on squared paper, directly experiencing WHY the perpendicular height is essential and WHY its absence in the real shamba creates the central puzzle that will drive the rest of the unit.
Safety Notes	No significant safety concerns for this lesson. Ensure rulers and pencils are handled responsibly. If scissors are used to cut squared paper triangles, remind learners to handle them carefully and pass them handle-first.

B. LESSON OVERVIEW

Learners are introduced to the anchoring phenomenon of a Nakuru land surveyor tasked with finding the exact area of an irregular five-sided shamba. Through a whole-class phenomenon launch, individual prediction, hands-on measurement of triangles on squared paper, and a structured class discussion, learners revisit the Area = ½ × base × height formula and — crucially — encounter its limitation: it requires a perpendicular height that is not always available. The lesson closes with the creation of the unit's Driving Question Board, anchoring all future learning to the surveyor's challenge. This lesson covers NGSS Science and Engineering Practices: Asking Questions, Analysing and Interpreting Data, and Constructing Explanations.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane]	Learners are shown a hand-drawn sketch of an irregular five-sided shamba in Nakuru County (displayed on the board or printed on a half-sheet). The sketch shows	1. Before revealing the sketch, say: 'Before I show you anything, I want you to think about this — what do you already know about calculating area of shapes?' Cold-	Strategy: Anchored Prediction (Elicit-Before-Instruction). By asking learners to commit a written prediction to paper BEFORE instruction begins, the teacher	Observable indicator: Teacher scans written predictions during circulation — look for whether learners (a) identify that the shape must be broken into simpler parts,

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Logarithm Triangle (PLIX) Source: CK-12 🔍 Search ARES for similar readings	five boundary fence posts labelled A, B, C, D, E with unequal side lengths written alongside (e.g., AB = 48 m, BC = 35 m, CD = 52 m, DE = 40 m, EA = 37 m) but NO perpendicular heights marked. Learners are told: 'Three siblings — Achieng, Baraka, and Cherotich — have inherited this shamba from their late father. A land surveyor from Nakuru town has been hired to calculate the exact area so the land can be fairly divided. Your task right now: In your exercise book, write down EXACTLY how you would calculate the area of this plot using only a tape measure and what you already know about area.' Learners work individually and silently for 4 minutes, then share predictions with a shoulder partner for 2 minutes.	call 3 learners for quick responses. Write key words (rectangle, length × width, triangle, base, height) on the board. 2. Reveal the shamba sketch. Say: 'This is a real problem facing land surveyors across Kenya every day — in Nakuru, in Eldoret, in Kisumu. The shamba belongs to a real family. The surveyor is standing right here at fence post A. What does she do?' 3. Give the written prediction task. Circulate silently — do NOT answer questions. After 4 minutes, say: 'Now share your prediction with your partner. You have 2 minutes. Go.' 4. After partner share, cold-call 4 learners to share predictions. Write a short phrase from each on the board under the heading 'OUR PREDICTIONS'. Do not correct — affirm all contributions: 'Interesting. Let's keep that idea on the board and return to it.' 5. Ask the class: 'How many of you mentioned a formula that needs a HEIGHT? Put your hand up.' Pause for 10 seconds. Then say: 'That height — that is going to be the most important word in this entire unit. Remember it.'	activates prior knowledge, surfaces misconceptions (e.g., 'just multiply all sides', 'use length × width'), and creates a personal stake in resolving the puzzle. Displaying predictions on the board creates a public record that will be revisited and revised across the unit, building metacognitive awareness of how understanding grows.	(b) correctly recall that triangle area involves a height, (c) show confusion about how to get the height. Use this to gauge the class's starting point. Cold-call responses reveal whether learners can connect 'area' to a formula, not just a vague process. Red flag: If fewer than 50% of learners mention height or any formula, spend an additional 3 minutes on a quick whole-class recall of area formulas before proceeding.
Observe Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Logarithm Triangle (PLIX) Source: CK-12	Learners receive a printed or hand-drawn A4 worksheet containing three triangles drawn on a 1 cm squared grid paper background. Triangle 1: right-angled triangle (base = 6 cm, height = 4 cm — height is a grid line, easy to read). Triangle 2: acute triangle (base = 8 cm, a slanted height is drawn as a dotted line = 5 cm). Triangle 3: obtuse triangle (base = 10 cm, NO height line drawn — learners must construct or estimate the perpendicular height themselves	1. Distribute worksheets and materials. Say: 'You have three triangles. For each one, I want you to do four things — I'll write them on the board.' Write the four steps clearly. 2. Circulate actively. At Triangle 1, prompt: 'How many full squares are inside? How many half-squares? What does that give you?' Wait 10 seconds before giving any further prompt. 3. At Triangle 2, ask pairs: 'Where exactly is the perpendicular height? How do you know it is perpendicular?' Cold-call 2 pairs to	Strategy: Concrete-Representational-Abstract (CRA) Progression with Purposeful Struggle. Learners begin with counting squares (concrete estimation), move to reading a drawn height (representational), and then must construct the height themselves (abstract application). Triangle 3 is deliberately designed to create productive struggle — the absence of a drawn height mirrors exactly the surveyor's dilemma with the real shamba. The comparison between counting	Observable indicator: Check that learners correctly identify the perpendicular (not slant) height for Triangle 2. Common error: measuring the slant side as the height. For Triangle 3, observe whether learners use the set square correctly to draw a perpendicular from the apex to the base extended. Ask targeted pairs: 'Show me exactly where your height starts and ends.' Learners should be able to state: 'The height must be perpendicular to the base.' If the majority measure slant

🔍 Search ARES for similar readings	using a ruler and set square). For each triangle, learners: (a) Count the full squares and half-squares inside the triangle to estimate area by counting. (b) Identify and measure the base and perpendicular height. (c) Apply the formula Area = ½ × base × height and record their answer. (d) Compare their counting estimate with their formula answer. Learners work in pairs. They also have access to a 30 cm ruler and a set square.	explain. 4. At Triangle 3 — this is the critical triangle. When learners struggle to find the height, do NOT immediately help. Say: 'This is the most important triangle on your sheet. Sit with the confusion for a moment. What would you NEED to draw to find the height?' Wait 15 seconds. Then hint: 'Think about what perpendicular means. Use your set square.' 5. After 15 minutes of pair work, bring the class together. Ask: 'For Triangle 3 — who found the height easily? Who struggled?' Hands up, scan, then cold-call one learner from each group. 6. Say: 'Now look back at our surveyor's shamba sketch. Does the shamba have any perpendicular heights drawn? What does that mean for our formula?'	estimate and formula answer builds trust in the formula while making the HEIGHT visible as the critical variable.	height for Triangle 2, pause the class and address this misconception whole-class before moving to Triangle 3.
Explain Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Logarithm Triangle (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners now connect their hands-on findings back to the Nakuru shamba phenomenon. Working in groups of four, they receive a 'Surveyor's Challenge Card' that reads: 'Surveyor Wanjiku is standing at the shamba in Nakuru. She has measured all five sides of the plot. She wants to split the plot into triangles to calculate the total area. She draws diagonal lines from corner A to corner C, and from corner A to corner D, creating three triangles: Triangle ABC, Triangle ACD, and Triangle ADE. She knows all side lengths but she CANNOT measure any perpendicular heights because the ground is uneven and she cannot stretch a tape measure at a right angle across the maize crop growing in the shamba. She asks you: Can I still use Area = ½ × base × height? What is the problem? What do I need instead?'	1. Distribute Surveyor's Challenge Cards and manila paper. Say: 'Surveyor Wanjiku needs your help. In your group, answer her question — and be specific. Don't just say the formula won't work. Explain WHY.' 2. Circulate during group work. Listen for groups that identify the perpendicular height as the missing element. Prompt groups that are stuck: 'What does ½ × base × height actually need from you — what must you physically measure or know?' Wait 10 seconds. 3. During gallery walk (5 minutes), prompt learners to write sticky notes that say either 'We agree because…' or 'We have a question about…'. 4. After gallery walk, bring class back together. Ask: 'What was the most common problem all groups identified?' Cold-call 3 learners. Establish the class consensus statement: 'To use Area = ½ ×	Strategy: Collaborative Argumentation via Gallery Walk. By requiring groups to write a position on the surveyor's problem and then critique each other's reasoning through sticky notes, learners practise constructing and evaluating explanations — a core Science and Engineering Practice. The gallery walk externalises thinking and creates a low-stakes peer-review environment. The teacher's final consolidation move (the class consensus statement in the box) converts the collaborative discussion into a shared, recorded claim that anchors the unit's knowledge-building trajectory.	Observable indicator: Group manila paper responses should clearly state that the perpendicular height is the missing element and that the formula cannot be applied without it. Look for groups that also begin to speculate about alternative methods (e.g., 'maybe we use angles' or 'maybe there is another formula') — these are high-value ideas to affirm publicly. Red flag: Groups that say 'she can just guess the height' or 'she can measure diagonally' need targeted questioning: 'What does perpendicular mean? Is a diagonal measurement the same as a perpendicular height?'

	Groups discuss for 5 minutes and write a group response on a piece of manila paper. Groups then share responses in a 'gallery walk' — each group's paper is posted on the wall and learners circulate with sticky notes to add comments or questions.	base × height, we MUST know the perpendicular height.' Write this on the board in a box labelled 'WHAT WE NOW KNOW'. 5. Then say: 'So here is our big question — if Surveyor Wanjiku does NOT have the perpendicular height, but she DOES have the lengths of all three sides of each triangle, is there another way? Take 10 seconds and think silently.' Pause for 10 seconds. Say: 'That question — that is what this unit is about. We will answer it together over the next lessons.' 6. Return to the shamba sketch. Say: 'By the end of this unit, you will be able to calculate the exact area of this shamba. Today, you have taken the first step.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Logarithm Triangle (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives three sticky notes. They are asked to write one item per sticky note: (1) BLUE sticky note — one thing they NOW KNOW about area of triangles and polygons (from today's lesson). (2) YELLOW sticky note — one question they STILL HAVE about how to find area when the height is unknown. (3) GREEN sticky note — one prediction or idea about what mathematics might help the surveyor solve her problem. Learners write individually (3 minutes), then come to the board one row at a time and place their sticky notes in the correct column of the class Driving Question Board. The DQB has three columns: 'What We Know', 'What We Wonder', 'What We Think Will Help'. The teacher reads aloud a selection of yellow (wonder) sticky notes and circles two or three that will explicitly be answered in upcoming lessons.	1. Introduce the DQB on a large section of the classroom wall or a dedicated display board. Say: 'This board is going to track our journey as a class. By the last lesson of this unit, every single yellow sticky note — every question you write today — should have an answer on this board. That is our goal.' 2. Distribute sticky notes. Say: 'Blue: one thing you KNOW. Yellow: one thing you WONDER. Green: one idea you PREDICT. Write your name small on the back of each note. You have 3 minutes. Silent, individual thinking.' Wait 3 minutes without circulating. 3. Invite learners row by row to place their notes. As they do, read a few aloud and narrate: 'Someone wonders whether there is a formula that uses only sides and no height — excellent. Let's put that right here.' 4. After all notes are placed, stand back and say: 'Look at this board. This is every question that this class has about	Strategy: Driving Question Board (DQB) as a Living Knowledge Artefact. The DQB is not a one-time activity — it is a cumulative display that will be revisited at the start and end of every lesson in the unit. By having learners categorise their thinking into 'Know / Wonder / Predict', the teacher makes metacognition visible and communal. Circling specific questions that future lessons will answer creates a narrative thread — learners can see their questions being systematically resolved, which builds motivation and a sense of intellectual progress tied directly to the phenomenon.	Observable indicator: Yellow sticky notes should contain genuine conceptual questions (e.g., 'How do you find area without height?', 'Can you use angles instead?', 'What if the triangle is very flat?') rather than procedural ones (e.g., 'What is the formula?'). If most yellow notes are purely procedural, the phenomenon has not yet created sufficient cognitive dissonance — consider spending 2 additional minutes re-narrating the surveyor's dilemma more vividly before the next lesson. Green prediction notes that mention angles, sine, or 'a different formula' indicate learners who are ready for acceleration in Lesson 2.

		area of polygons right now. These questions are not weaknesses — they are the engine of our learning.' 5. Circle 2–3 yellow notes that align with upcoming lessons (Heron's formula, sine rule for area). Say: 'These questions — right here — we will answer in Lesson 2 and Lesson 3. Keep them in mind.' 6. Take a photo of the DQB to display digitally or keep for reference in future lessons.		
Model Building Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Logarithm Triangle (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners individually draw their first 'Area Model' in a dedicated section of their exercise book labelled 'MY GROWING MODEL — Area of Polygons'. For Lesson 1, the model has three components: (a) A sketch of the Nakuru shamba divided into three triangles (copied from the board diagram used in Phase 3). (b) The formula Area = ½ × base × height written clearly, with the word HEIGHT underlined and annotated: 'This MUST be perpendicular to the base.' (c) A 'What I Still Need' box where learners write in their own words: 'To find the area of Triangle ABC in Surveyor Wanjiku's shamba, I still need to know _______ because _______.' Learners complete the sentence stem individually. Three learners are cold-called to share their completed sentence aloud.	1. Say: 'In your exercise book, find a clean page and write the heading: MY GROWING MODEL — Area of Polygons, Lesson 1. This model will GROW every lesson. By Lesson 8, it will show you everything you need to solve the surveyor's problem completely.' 2. Guide learners step by step: 'First, sketch the shamba and its three triangles. Use my diagram on the board.' Give 2 minutes. 'Now write the formula. Underline HEIGHT and write your annotation.' Give 1 minute. 'Now complete the sentence stem in the What I Still Need box. Be specific — do not just say you need more information. Say exactly what mathematical information is missing.' Give 2 minutes. 3. After 5 minutes of model-building, cold-call 3 learners to read their completed sentence stem aloud. After each response, ask the class: 'Does everyone agree with that? Thumbs up, thumbs sideways, or thumbs down.' Scan the room. 4. Close the model phase by saying: 'Your model today is incomplete — and that is completely correct. It SHOULD be incomplete. In the next lesson, we will add a new tool to your model	Strategy: Progressive Model Revision (PMR). The 'Growing Model' is a cumulative knowledge-representation tool in which learners literally add new content to the same diagram each lesson. This approach, drawn from NGSS Modelling practices, makes the growth of understanding visible and tangible. The 'What I Still Need' sentence stem is a deliberate incompleteness prompt — it keeps the phenomenon's unresolved question alive at the individual level and creates a personal 'knowledge gap' that motivates the next lesson. The cold-call sharing makes the model a semi-public artefact, building classroom norms around intellectual vulnerability and growth.	Observable indicator: The 'What I Still Need' sentence stem should specifically name the perpendicular height as the missing element (e.g., 'I still need to know the perpendicular height of each triangle because the formula ½ × base × height cannot be used without it'). Learners who write vague responses (e.g., 'I need more measurements') should be prompted: 'Which specific measurement? What does that measurement look like on your diagram?' Exit check: As learners leave, teacher stands at the door and asks each learner to say one word that describes what the formula ½ × base × height absolutely requires. Correct answer: 'height' or 'perpendicular'. This takes 15 seconds per learner and provides a rapid whole-class data point.

		that begins to solve Surveyor Wanjiku's problem even without a perpendicular height. Come ready.'		

D. TEACHER REFLECTION
After delivering this lesson, reflect on the following questions: (1) Did the phenomenon — the Nakuru shamba and Surveyor Wanjiku — genuinely create curiosity and cognitive dissonance, or did learners treat it as a routine word problem? If the latter, how can the narrative be made more vivid or personal in the next lesson? (2) During Phase 2, did the majority of learners correctly distinguish between perpendicular height and slant height for Triangle 2? If not, this misconception must be explicitly re-addressed at the start of Lesson 2 before introducing Heron's formula. (3) How rich were the yellow (wonder) sticky notes on the DQB? Did learners ask questions that pointed toward trigonometry and alternative formulas, or were their questions still at the level of basic arithmetic? The sophistication of these questions will guide pacing decisions for Lessons 2 and 3. (4) Were all learners able to complete the 'What I Still Need' sentence stem with sufficient specificity? Learners who struggled with this may need a structured language scaffold (a more detailed sentence frame) in future model-building phases. (5) Were learners from all groups — including girls and learners who typically disengage — cold-called and included in public sharing? Review your cold-call record and adjust targeting in Lesson 2 to ensure equitable participation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a sketch of an irregular five-sided shamba in Nakuru County with five unequal side lengths and no perpendicular heights marked. They observed three triangles on squared paper — one with an easy right-angle height, one with a drawn perpendicular height, and one with NO height drawn — and used counting and the formula Area = ½ × base × height to find each triangle's area.
What did I learn?	Learners (re)established that the formula for the area of a triangle is Area = ½ × base × height, and that the height MUST be the perpendicular distance from the base to the opposite vertex. They learned that when a triangle is drawn in a non-right-angle orientation, the perpendicular height must be constructed — it is not the same as any side length. They connected this requirement directly to the surveyor's problem: the shamba cannot be measured with this formula alone because no perpendicular heights are accessible.
How does this explain the phenomenon?	Learners explained (in groups and in their individual Growing Model) that Surveyor Wanjiku cannot directly apply the formula Area = ½ × base × height to the triangles in the shamba because the perpendicular heights are not measurable on uneven ground with maize crops. They articulated this as a precise knowledge gap: to calculate the area of an irregular polygon by decomposing it into triangles, a method for finding triangle area WITHOUT a perpendicular height is needed. This gap launches the unit's investigation into Heron's formula and the sine area rule.

LESSON 2 (80 minutes): Area of a Triangle Using the Sine Formula (½ab sin θ)	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to calculate the area of a triangle using the sine formula (A = ½ab sin θ) when the perpendicular height is not directly measurable, and to apply this formula to real-world land surveying contexts in Kenya.
Knowledge	By the end of the lesson, the learner should be able to derive and state the sine area formula A = ½ab sin θ, explaining why it is needed when the perpendicular height of a triangle is unknown.
Skills	By the end of the lesson, the learner should be able to correctly apply A = ½ab sin θ to calculate the area of triangles given two sides and an included angle, including triangular sections of irregularly shaped farm plots.
Attitudes	By the end of the lesson, the learner should appreciate the elegance and practical utility of the sine formula as a tool that empowers land surveyors and farmers in Kenya to solve real measurement problems without physical access to every dimension.
Key Inquiry Question	How do we determine the area of a polygon? How do we apply area of polygons in real-life situations?
Purpose in Storyline	In Lesson 1, students established that the surveyor's shamba in Nakuru cannot be measured using ½ × base × height alone because no perpendicular height is available for the triangular sections. Lesson 2 directly resolves this first obstacle: students derive the sine formula and use it to calculate the area of a triangular corner of the shamba, advancing the class's ability to answer the driving question. This is the second piece of evidence in the sequence — the sine formula becomes the surveyor's first reliable tool.
Safety Notes	No physical hazards anticipated. Ensure learners handle calculators and mathematical tables responsibly. Remind learners to work in degrees mode on their calculators when evaluating trigonometric ratios.

B. LESSON OVERVIEW
Learners begin by revisiting the Driving Question Board (DQB) from Lesson 1, recalling that the Nakuru surveyor cannot use ½ × base × height without a perpendicular height. Through a guided derivation activity, learners discover that expressing height in terms of a known side and included angle (h = b sin θ) transforms the standard formula into A = ½ab sin θ. Pairs then apply the formula to two triangular farm-plot scenarios drawn from Nakuru County land data. The lesson ends with learners updating their cumulative polygon-decomposition model of the shamba and adding a new card to the DQB: "We can find area without height if we know two sides and the angle between them."

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Trigonometry Word Problems Principles - Basic] Source: CK-12	Learners are shown a scale diagram of a triangular corner of the Nakuru shamba (Triangle ABC: AB = 120 m, AC = 95 m, angle A = 68°). They are told the surveyor cannot physically	Display the triangular shamba diagram on the board (or printed handout). Say verbatim: 'Look carefully at this triangular corner of the shamba in Nakuru. The surveyor knows two side lengths	Prediction Journaling — learners activate prior knowledge of the ½ × base × height formula and surface their conceptual gap (no height available). Recording predictions in writing creates a personal	Observable indicator: Can the learner articulate that the missing perpendicular height is the specific obstacle? Listen for phrases such as 'I need the height from B to AC' or 'the formula needs h but I don't

🔍 Search ARES for similar videos 📖 **HTML:** Determine and Use the Sine Ratio (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	measure the height from B perpendicular to AC because a seasonal stream runs through that line. Working individually in their exercise books for 4 minutes, learners write: (a) their estimate of the area of the triangle, (b) what information they wish they had, and (c) a prediction of whether the standard ½ × base × height formula can be rescued or a completely new formula is needed. Learners then share predictions with a shoulder partner for 2 minutes before three predictions are cold-called to the class.	and the angle between them — but the stream prevents measuring the height. In your book, write your estimate of the area, what information you wish you had, and whether you think the standard formula can be saved or must be replaced. You have 4 minutes — work in silence.' Set a visible 4-minute timer. After 4 minutes, say: 'Share your prediction with your partner — 2 minutes.' After sharing, cold-call exactly 3 learners using name sticks (one strong, one mid, one quiet learner). For each response, say: 'Interesting — keep that prediction in mind as we work today.' Do NOT confirm or correct predictions yet. Write the three predictions verbatim on the corner of the board under the heading 'Our Predictions.'	benchmark that learners will revisit during the Explain phase, producing cognitive dissonance that motivates the derivation.	have it.' Learners who cannot identify the specific gap will need targeted scaffolding during the derivation.
Observe Phase 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Determine and Use the Sine Ratio (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work in pairs through a structured derivation worksheet (or teacher-guided board work). Step 1 — Recall: pairs write the standard area formula A = ½ × base × height for triangle ABC with base b = AC. Step 2 — Express h in terms of what we know: using the right-angled triangle formed by dropping a perpendicular h from B to AC, pairs use trigonometry (sin θ = opposite/hypotenuse) to write h = c sin A (where c = AB). Step 3 — Substitute: pairs substitute h = c sin A into A = ½ × b × h to obtain A = ½ × b × c × sin A, i.e., A = ½ bc sin A. Step 4 — Generalise: pairs write the formula in its general form A = ½ab sin θ where a and b are the two known sides and θ is the included angle. Step 5 — Verify with the shamba triangle: pairs calculate A = ½ × 120 × 95 × sin 68° using calculators, obtaining	Distribute the derivation worksheet. Say verbatim: 'We are going to rescue the formula together — step by step. Work in pairs. Follow Steps 1 to 4 on your sheet. When you reach Step 5, use your calculator to compute the area. You have 15 minutes.' Circulate and check: at Step 2, pause the class after 5 minutes and ask: 'Which trigonometric ratio connects the height h, the hypotenuse AB, and the angle A?' Wait 12 seconds. Cold-call 2 pairs. If pairs are stuck, prompt: 'Draw the right-angled triangle inside the bigger triangle — label the opposite and hypotenuse.' At Step 5, ask: 'Before you type into the calculator, what mode must it be in?' Wait 10 seconds. Cold-call 1 learner. After pairs complete Step 5, say: 'Compare your answer with another pair — did you get	Guided Algebraic Derivation with Immediate Application — rather than presenting the formula as a given rule, learners construct it from the standard formula using known trigonometry. This derivation-first approach (an NGSS Science and Engineering Practice: Using Mathematics and Computational Thinking) ensures learners understand why the formula works, not just how to use it. Immediate application to the shamba context anchors the abstract algebra in the surveyor's real problem.	Observable indicator: At Step 2, the learner correctly writes h = c sin A (or equivalent). At Step 5, the learner obtains 5 285–5 295 m² (accept rounding variation). At the board example, the learner correctly identifies both known sides and the included angle before substituting. Flag any learner who uses a non-included angle — this is the most common error and must be addressed before the Explain phase.

	approximately 5 290 m². Teacher then works through a second Nakuru-context example on the board: a triangular plot near Nakuru town with sides 85 m and 110 m and included angle 52°, narrating each step aloud.	approximately 5 290 square metres?' Then move to the second board example, narrating: 'The surveyor now moves to a second triangular section of the same shamba near Nakuru town — 85 m, 110 m, angle 52°. Watch how I apply our new formula.' Work through it step by step, pausing to ask: 'What do I substitute for a and b here?' Wait 10 seconds. Cold-call 1 learner. Emphasise: 'The angle MUST be the included angle — the one sitting between the two known sides.'		
Explain Phase 📹 **VIDEO:** Trigonometry Word Problems Principles - Basic Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Determine and Use the Sine Ratio (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually solve two structured problems that escalate in complexity, both set in Kenyan land contexts. Problem 1 (Direct Application): A triangular shamba in Kericho has two sides of 74 m and 88 m with an included angle of 75°. Calculate its area. Problem 2 (Analytical): A surveyor measures a triangular plot outside Kisumu and records sides of 60 m and 90 m. She records the included angle as 30°. Her assistant says the area must be larger than a nearby rectangular plot measuring 40 m × 50 m (area 2 000 m²). Is the assistant correct? Show all working and justify. After solving individually (10 minutes), learners compare answers in groups of four, then a class discussion unpacks Problem 2 — where learners confront the idea that sin 30° = 0.5 yields a surprisingly small area (1 350 m²), smaller than the rectangle, teaching them that the included angle size critically affects area. Teacher closes by explicitly connecting back: 'Return to the Nakuru shamba. The surveyor has now solved the first triangular corner. Which of the	Say verbatim: 'Close your derivation sheet. In your books, solve both problems individually — you have 10 minutes. Show every step of substitution.' Set a 10-minute timer. Circulate — for learners who are stuck on Problem 2, prompt: 'What is sin 30°? Substitute that value first, then multiply.' After 10 minutes, say: 'Compare answers in your group of four — do you all agree? You have 3 minutes.' Cold-call one group to present Problem 2 on the board. After the presentation, pose to the whole class: 'Why did the assistant get it wrong? What did he forget to consider?' Wait 15 seconds. Cold-call 3 learners. Guide the class to articulate: 'Area with the sine formula also depends heavily on the angle — a small angle gives a small area even with long sides.' Then say: 'Now look at the Nakuru shamba diagram. Point to a corner where you could use today's formula.' Wait 10 seconds. Cold-call 2 learners to come to the board and point. Affirm correct corners and note: 'This is exactly the tool the surveyor needed.'	Graduated Problem Solving with Conceptual Trap (NGSS Practice: Constructing Explanations) — Problem 1 confirms procedural fluency; Problem 2 is deliberately designed to challenge the intuition that larger sides always mean larger area, forcing learners to reason about the role of the included angle. The class discussion of the 'trap' deepens conceptual understanding beyond rote formula use. Connecting back to the shamba diagram reinforces the storyline thread.	Observable indicator: Problem 1 correct answer is approximately 3 148 m² (accept 3 140–3 155 m²). Problem 2 correct area is 1 350 m², and the learner explicitly states the assistant is wrong, justifying with the computed value. Learners who cannot identify an applicable corner on the shamba diagram likely have not yet connected the formula to the polygon-decomposition strategy — note these learners for targeted support in Lesson 3.

	five-sided plot's corners could she tackle next using this formula?'			
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Trigonometry Word Problems Principles - Basic] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Determine and Use the Sine Ratio (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes one sticky note (or index card) responding to the prompt: 'What new evidence from today's lesson helps the Nakuru surveyor answer the driving question — how can she calculate the exact area of the irregularly shaped shamba?' Learners also write one remaining question they still have. Cards are placed on the DQB under two columns: 'Evidence We Now Have' and 'Questions We Still Have.' The teacher reads three evidence cards and two question cards aloud to the class, annotating the DQB. The class agrees on a collective evidence statement to add: 'When two sides and the included angle of a triangle are known, $A = \frac{1}{2}ab \sin \theta$ gives the exact area — even without a measurable height.' The teacher also returns to the three predictions written at the start of the lesson and asks: 'Were any of our predictions correct? What surprised you?'	Distribute sticky notes or index cards. Say verbatim: 'Write one piece of evidence from today that helps the surveyor, and one question you still have — you have 3 minutes.' After collection, read three evidence cards aloud verbatim, then ask: 'Does everyone agree this is evidence for our driving question? Thumbs up, sideways, or down.' Read two question cards aloud and say: 'These are great questions — hold them. Some will be answered in Lessons 3 and 4.' Write the class consensus evidence statement on the DQB in a different colour marker. Then point to the Predictions corner: 'At the start of the lesson, [Learner Name] predicted the formula could be rescued. Were they right? How?' Wait 10 seconds. Cold-call the learner whose prediction was referenced. Affirm: 'Yes — and the rescue came from trigonometry we already knew.'	Driving Question Board — Evidence Tracking (NGSS Practice: Engaging in Argument from Evidence): the DQB makes the cumulative nature of the evidence sequence visible. By physically placing a new card each lesson, learners see their growing toolkit. The return to opening predictions reinforces metacognitive reflection and shows learners that productive scientific thinking often starts with good predictions, not correct ones.	Observable indicator: The learner's evidence card correctly names the formula $A = \frac{1}{2}ab \sin \theta$ and specifies the condition (two sides and included angle known). A strong card also links to the shamba context (e.g., 'The surveyor can now find the area of the triangular corners of the five-sided plot'). Vague cards ('we learned a new formula') indicate surface-level engagement — these learners should be paired with stronger peers during Lesson 3's group work.
Model Building Phase 📹 **VIDEO:** [Trigonometry Word Problems Principles - Basic] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Determine and Use the Sine Ratio (Flexbook)] Source: CK-12 🔍 Search ARES	Learners retrieve their cumulative shamba diagram from Lesson 1 (a five-sided irregular polygon with labelled boundary posts). Using the two given measurements for Triangle ABC (AB = 120 m, AC = 95 m, angle A = 68°), learners shade Triangle ABC on their diagram, write the calculated area (≈ 5 290 m^2) inside the shaded region, and annotate: 'Area found using $A = \frac{1}{2}ab \sin \theta$.' Learners then write a sentence below their model: 'The sine formula solves the surveyor's problem for Triangle ABC. We still need to find areas	Say verbatim: 'Take out your shamba diagram from Lesson 1. Shade Triangle ABC, write the area you calculated inside it, and annotate the formula you used. You have 5 minutes.' Set a timer. Circulate and look for: correct shading of Triangle ABC only (not the whole polygon), correct area value, correct formula annotation. After 5 minutes, ask: 'Hold up your diagram — show your partner where you shaded.' Then ask the whole class: 'How much of the shamba have we calculated so far?' Accept responses. Say: 'One	Cumulative Model Revision (NGSS Practice: Developing and Using Models): the shamba diagram functions as a living, lesson-by-lesson model. Each lesson, one more region is shaded and labelled, making the decomposition strategy concrete and progressive. Learners can literally see how much of the problem has been solved and how much remains, sustaining motivation and purpose across the eight-lesson unit.	Observable indicator: The learner's diagram shows Triangle ABC correctly shaded with area ≈ 5 290 m^2 and the annotation '$A = \frac{1}{2}ab \sin \theta$.' The sentence below the model explicitly names both what has been solved and what remains. Extension learners correctly identify at least one additional triangular section that could use the sine formula, demonstrating readiness for Lesson 3's Heron's formula context.

for similar readings	for the remaining sections of the shamba.' Learners who finish early identify which other triangular sections of the shamba also have two sides and an included angle labelled, and predict which can also be solved with today's formula.	corner. What must we do to find the total area?' Cold-call 2 learners. Guide toward: 'We need to decompose the whole polygon into triangles and find each one's area.' Close with: 'In Lesson 3, we will meet Heron's formula — a second tool for triangles where we know three sides but no angle. Together, these two tools will let the surveyor crack the entire shamba.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) During the derivation, were learners able to independently write h = c sin A at Step 2, or did most pairs need the prompt about the right-angled triangle? If more than one-third of the class needed the prompt, plan a brief 5-minute trigonometry ratio recap at the start of Lesson 3. (2) In Problem 2, did learners successfully identify that the assistant was wrong, and could they articulate why (the role of the included angle)? If fewer than half the class reasoned correctly about the angle's effect, build a paired comparison activity into Lesson 3's warm-up where learners compare two triangles with the same sides but different included angles. (3) Review the DQB evidence cards — are the 'Questions We Still Have' pointing toward Heron's formula (e.g., 'What if we don't have any angle at all?') or toward polygon decomposition (e.g., 'How do we add up all the triangle areas?')? Use the distribution of questions to decide whether to spend more time on the derivation or the application in Lesson 3. (4) Were learners correctly annotating their shamba models, and does each learner have a diagram they can build on? Any learner without a usable diagram must be given a clean copy before Lesson 3 begins.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed a scale diagram of a triangular corner of the Nakuru shamba (AB = 120 m, AC = 95 m, angle A = 68°) where a seasonal stream blocked direct measurement of the perpendicular height. They followed a step-by-step derivation showing how the perpendicular height h can be expressed as h = c sin A using trigonometry, transforming the standard formula into A = ½ab sin θ. They also observed that in Problem 2, a small included angle (30°) produced a surprisingly small area — smaller than a nearby rectangle — even with sides of 60 m and 90 m.
What did I learn?	Learners learned that when the perpendicular height of a triangle is not directly measurable but two sides and the included angle are known, the area can be calculated exactly using A = ½ab sin θ. They learned that the angle in this formula must be the included angle — the angle formed between the two known sides — and that the size of this angle critically determines the area. They also learned that this formula is derived directly from the standard ½ × base × height formula using the trigonometric identity h = b sin θ, meaning no new mathematics is invented — only existing tools are combined in a new way.
How does this explain the phenomenon?	Learners explained that the Nakuru surveyor can now calculate the area of any triangular section of the five-sided shamba where two side lengths and the included angle are recorded in her field notes, even if she could not physically measure the height due to the stream. They explained why the assistant in Problem 2 was wrong — longer sides do not guarantee a larger area if the included angle is small. They updated their cumulative shamba model by shading and labelling Triangle ABC (area ≈ 5 290 m^2) and articulated that the sine formula is the first of multiple tools needed to decompose and solve the entire irregular polygon.

LESSON 3 (80 minutes): Applying the Sine Formula for Area of a Triangle	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To develop learners' ability to calculate the area of a triangle using the sine formula (A = ½ab sinC) and to select appropriately between the sine formula and the base-height formula based on given information.
Knowledge	Learners will know that the area of a triangle can be calculated using two sides and the included angle via A = ½ab sinC, and understand when this formula is necessary as opposed to A = ½ × base × height.
Skills	Learners will be able to: (a) identify the two sides and included angle in a given triangle; (b) apply the sine formula A = ½ab sinC correctly; (c) select between ½bh and ½ab sinC based on the information provided; (d) solve real-life area problems set in Kenyan contexts.
Attitudes	Learners will appreciate the power of the sine formula as a practical tool for land surveyors, builders, and farmers who cannot always measure perpendicular heights directly.
Key Inquiry Question	How do we determine the area of a polygon? How do we apply area of polygons in real-life situations?
Purpose in Storyline	In Lesson 1 learners identified that the Nakuru surveyor cannot always find a perpendicular height. In Lesson 2 they explored Heron's formula for when all three sides are known. Now in Lesson 3, learners discover the sine formula — the surveyor's tool when two sides and the included angle are known — and practise selecting the right formula. This directly advances their ability to decompose the five-sided shamba into triangles and calculate each triangle's area.
Safety Notes	No physical safety hazards. Remind learners to use calculators responsibly and verify that their calculator is set to DEGREE mode before computing sine values. Emotional sensitivity: the anchoring phenomenon involves a family inheritance after a father's passing — treat references to this context with respect.

B. LESSON OVERVIEW

Learners revisit the Nakuru shamba phenomenon and recall that the surveyor cannot always measure a perpendicular height. Building on Lesson 2 (Heron's formula — all three sides known), learners now explore the sine formula A = ½ab sinC for when two sides and the included angle are known. They watch a short narrated video showing the sine formula applied to a triangular roof panel in Eldoret and a boat sail on Lake Victoria, then work through guided and independent Kenyan-context examples. The lesson closes with a formula-selection card sort and a DQB update that records how the surveyor's toolkit is growing. By the end, learners can confidently choose between ½bh and ½ab sinC and apply the sine formula accurately.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane]	Learners are shown a diagram of a triangle with two sides labelled (a = 9 m, b = 12 m) and the included angle labelled (C = 65°) but NO perpendicular height given. The	Display the triangle diagram on the board/screen. Say: 'Look carefully — you have two sides and the angle squeezed between them. You do NOT have the height. In	Predict-then-Revisit: By committing predictions to writing before instruction, learners activate prior knowledge (½bh) and create a cognitive gap when	Observable indicator: Every learner has a written prediction in their book. Teacher scans during partner share — look for learners who write 'impossible without

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Determine and Use the Sine Ratio (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	prompt on screen reads: 'The Nakuru surveyor has split the shamba into triangles. For this triangle she knows two sides and the angle between them — but she cannot physically measure the height. Do you think we can still find the area? How?' Learners write a silent individual prediction in their exercise books (3 sentences minimum), then share with a shoulder partner for 90 seconds.	your books, write your prediction: can we find the area? If yes, how might we do it? If no, why not? You have 2 minutes — silent, individual thinking first.' Start timer. After 2 minutes, say: 'Now share your prediction with your partner. You have 90 seconds.' After partner share, cold-call 3 pairs (select one confident, one mid-level, one uncertain learner): 'Akinyi, what did you and your partner predict?' Record key prediction words on the board in a column labelled BEFORE. Do NOT confirm or deny — say: 'Interesting — let us keep these predictions open and gather some evidence.'	they realise height is missing. This gap motivates genuine attention during the observe phase.	height' (reveals the productive confusion to be resolved) versus learners who guess 'use sine' (reveals prior exposure). Note these two groups to differentiate cold-call questioning in later phases.
Observe Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Determine and Use the Sine Ratio (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners watch a 7-minute narrated instructional video (ARES platform) in two segments. SEGMENT 1 (3.5 min): A Kenyan builder in Eldoret calculates the area of a triangular roof panel with sides 6 m and 8 m and included angle 50°. The video derives A = ½ab sinC step by step, showing where sinC comes from geometrically (the height h = a sinC). SEGMENT 2 (3.5 min): A fisherman on Lake Victoria estimates the canvas area of a triangular boat sail with sides 2.5 m and 3.2 m and included angle 72°, using the same formula. After each segment, learners complete a structured observation notes table in their books with columns: 'What information was given | Formula used | Steps taken | Answer with units'.	Before playing: 'As you watch, fill in your observation table for each example. Pause mentally at each step — do NOT just copy the answer, write down WHY each step happens.' Play Segment 1. PAUSE video after the derivation of h = a sinC. Say: 'Stop — turn to your partner. In one sentence, explain why h equals a times sinC. You have 15 seconds. [WAIT 15 seconds] Baraka, tell me what your partner said.' Resume and complete Segment 1. Pause again before Segment 2: 'Before we see the sail example, predict the answer for the sail — write it in your table now. [WAIT 10 seconds]' Play Segment 2. After video ends: cold-call 2 learners to read their completed observation table aloud, asking the class 'Do you agree with their steps? Thumbs up or thumbs sideways.'	Segmented Video with Embedded Stops: Breaking the video into segments with structured observation tables and mid-video predictions forces active processing rather than passive viewing. The geometric derivation (h = a sinC) connects the new formula to the known ½bh formula, building conceptual coherence rather than rote memorisation.	Observable indicator: Learners' observation tables show all four columns populated with correct information for both examples. Teacher circulates during the pause after Segment 1 — check that learners have written h = a sinC (not just 'use sine'). Learners who cannot explain the derivation in one sentence need targeted support in Phase 3.
Explain Phase 📹 **VIDEO:** Parallelogram on the coordinate	Learners work through three tiered practice problems in pairs, all set in Kenyan contexts, moving from guided to independent: PROBLEM	Distribute the tiered worksheet. Say: 'Problem 1 is guided — use the template and work with your partner. You have 5 minutes.'	Tiered Worked Examples with Formula-Selection Justification: Moving from guided substitution to independent selection mirrors the	Observable indicators: (1) Problem 1 answer: A = ½ × 14 × 18 × sin72° ≈ 119.9 m² — check substitution and calculator use. (2)

plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Determine and Use the Sine Ratio (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	1 (Guided — Nakuru Shamba): One triangular section of the Nakuru shamba has sides of 14 m and 18 m with an included angle of 72°. Calculate its area. [Scaffolded template on worksheet — learners fill in a = ___, b = ___, C = ___, then substitute.] PROBLEM 2 (Semi-independent — Kericho Tea Farm): A triangular section of a Kericho tea farm has two boundary fences of 23 m and 31 m meeting at an angle of 58°. Find the area of this section to the nearest square metre. PROBLEM 3 (Formula Selection — Choice Task): Given three different triangles, each described differently (one with base and height, one with two sides and included angle, one with all three sides), learners must: (a) name the correct formula for each and (b) calculate the area of the triangle that uses the sine formula. Learners write a one-sentence justification for each formula choice.	Circulate — look for calculator errors (degree mode). After 5 min, cold-call one pair: 'Wanjiku, read your substitution step.' Write the correct working on the board. Say: 'Now Problem 2 — semi-independent, same process, no template. 6 minutes.' [WAIT, circulate — target pairs who struggled on Problem 1 with the prompt: 'Which two values are your sides? What is the angle between them?'] After 6 min, cold-call one mid-level learner: 'Otieno, what answer did you get and what units did you use?' Discuss units (m^2). Then: 'Problem 3 is the most important — you must JUSTIFY your formula choice. 8 minutes, still in pairs.' After 8 min, ask the whole class: 'For the triangle where you only know the base and height — which formula? [WAIT 10 seconds] Everyone write it on your mini-whiteboard / scrap paper — show me.' Scan responses. Cold-call 3 learners with different answers and resolve through class discussion. Close with: 'So the surveyor in Nakuru — which formula does she use for the triangles where she measured two sides and an angle? [WAIT 10 seconds] Chege?' Confirm: ½ab sinC.	surveyor's real decision-making process. Writing a justification sentence forces learners to articulate the condition under which each formula applies — a metacognitive move that prevents formula confusion in later lessons on polygons and Heron's formula.	Problem 2 answer: A = ½ × 23 × 31 × sin58° ≈ 303.0 m^2 — check rounding and units. (3) Problem 3 justification sentences correctly match formula to given information for all three triangles. Teacher flags any learner who uses ½bh on a problem where no height is given — this is the key misconception to address before Lesson 4.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Each learner writes one sticky note (or index card) and places it on the class Driving Question Board under one of three columns: 'WHAT WE NOW KNOW' / 'NEW QUESTIONS WE HAVE' / 'STILL NOT SURE ABOUT'. Learners are prompted with: 'Think about the Nakuru surveyor and her five-sided shamba. What can she now do that she could NOT do at the start of Lesson 1? What can she	Distribute sticky notes / index cards. Say: 'You have 3 minutes to write your contribution to the DQB. Think specifically about the surveyor's problem: she has a five-sided shamba and needs to find its total area. After today's lesson, what new tool does she have? What questions remain unanswered?' [WAIT 3 minutes — silent individual writing.] Direct learners to post their notes in the	Driving Question Board Update with Gallery Walk: The DQB makes the cumulative growth of the class's investigative toolkit visible. By explicitly linking today's new formula to the surveyor's unsolved problem, learners see the lesson not as an isolated topic but as one piece of a larger sensemaking journey. The gallery walk promotes peer-to-peer discourse and surfaces	Observable indicators: 'WHAT WE NOW KNOW' notes correctly reference the sine formula and its condition (two sides + included angle). 'NEW QUESTIONS' notes reveal whether learners are thinking ahead (e.g., 'What about when we only have three sides?' — confirms Heron's formula is retained from Lesson 2; or 'How do we add up all the triangle areas?' — shows readiness for polygon

📖 **HTML:** [Determine and Use the Sine Ratio (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	still NOT do?' The class then does a 3-minute gallery walk to read each other's sticky notes before returning to seats. The teacher reads out 2–3 selected notes aloud and links them explicitly to the driving question.	correct column. 'Stand up and do a gallery walk — read at least four other people's notes. You have 3 minutes.' After gallery walk, read aloud one 'WHAT WE NOW KNOW' note: 'Someone wrote: she can now calculate the area of each triangle if she knows two sides and the included angle. Is that accurate? [Cold-call 2 learners for agreement/disagreement.]' Read one 'NEW QUESTIONS WE HAVE' note — likely about irregular polygons or what to do when neither height nor angle is available. Say: 'Hold that question — it will be answered in Lessons 4 and 5.'	misconceptions publicly.	decomposition in Lesson 5). Teacher photographs the DQB at the end of class for tracking across the unit.
Model Building Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Determine and Use the Sine Ratio (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners return to their cumulative 'Surveyor's Toolkit' model — a visual diagram they have been building since Lesson 1. In Lesson 1, they added the phenomenon and the question. In Lesson 2, they added Heron's formula with the condition 'use when: all 3 sides known.' Today they add a new branch labelled 'Sine Formula: A = ½ab sinC' with the condition 'use when: 2 sides + included angle known.' They also add an arrow showing that the five-sided shamba can be decomposed into triangles, each of which might require EITHER Heron's OR the sine formula depending on what measurements the surveyor has. Learners annotate their model with one real example from today's lesson (e.g., the Eldoret roof panel or the Lake Victoria sail). As an exit task, each learner writes one sentence at the bottom of their model: 'The surveyor can now handle a triangle where ______ because ______.'	Say: 'Open your Surveyor's Toolkit model from Lesson 2. Today we add a new branch. Draw a new arm from the triangle node and label it: Sine Formula A = ½ab sinC. Write the condition next to it: USE WHEN — 2 sides and the included angle are known. Add your own example from today. You have 5 minutes.' Circulate — check that the condition is worded correctly and that learners distinguish it from Heron's condition. After 5 minutes: 'Now write your exit sentence at the bottom: The surveyor can now handle a triangle where BLANK because BLANK. Be specific — use the surveyor's shamba in your sentence.' [WAIT 3 minutes.] Cold-call 3 learners to read their exit sentence aloud. Confirm correct sentences, gently correct imprecise ones. Collect / photograph 5 models for teacher review. Close: 'The surveyor's toolkit is growing. In Lesson 4 we will look at what happens when the triangle is part of a larger	Cumulative Annotated Model (Surveyor's Toolkit): Adding to the same visual model across lessons makes conceptual growth tangible. The branching structure (Heron's vs. sine formula) reinforces that formula selection is conditional — a key mathematical reasoning skill. Connecting each new branch to the shamba phenomenon prevents isolated memorisation and keeps the inquiry coherent across all 8 lessons.	Observable indicators: (1) The new sine formula branch is correctly labelled with A = ½ab sinC and the correct condition. (2) The model includes an arrow showing polygon decomposition into triangles. (3) The exit sentence correctly identifies the condition for using the sine formula AND links it to the surveyor's shamba context. Teacher uses collected models to identify learners who still conflate Heron's and sine formula conditions — these learners receive a targeted prompt card at the start of Lesson 4.

		quadrilateral — a parallelogram or trapezium. How might the sine formula help there too?'		

D. TEACHER REFLECTION
After the lesson, reflect on the following: (1) Formula Confusion — Did any learners consistently apply ½bh when no height was given, or apply the sine formula when three sides were provided? These learners need a comparison table (Heron's vs. sine vs. base-height) before Lesson 4. (2) Calculator Errors — Were learners consistently in DEGREE mode? If more than 3 learners produced wrong answers due to radian mode, open Lesson 4 with a 2-minute calculator check routine. (3) DQB Quality — Were the 'NEW QUESTIONS' notes pointing forward (polygon decomposition, quadrilaterals) or backward (still confused about basic sine values)? If backward, allocate 10 minutes at the start of Lesson 4 for a sine values refresher. (4) Kenyan Context Engagement — Did learners engage more actively with the shamba / Eldoret roof / Lake Victoria examples than with abstract triangles? Note which contexts generated the most discussion — use these in Lessons 4 and 5. (5) Storyline Coherence — Can every learner now articulate at least one new thing the Nakuru surveyor can do that she could not do in Lesson 1? If not, the phenomenon link needs to be made more explicit in subsequent lessons.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners watched a narrated video showing a builder in Eldoret and a fisherman on Lake Victoria each calculate a triangle's area using the sine formula when no perpendicular height was available. They recorded structured observation notes for both examples.
What did I learn?	Learners learned that when two sides and the included angle of a triangle are known, the area can be calculated using $A = \frac{1}{2}ab \sin C$, because the perpendicular height can be expressed as $h = a \sin C$. They practised selecting between this formula and ½bh based on what information is given.
How does this explain the phenomenon?	Learners explained their formula selections in writing and connected the sine formula to the Nakuru surveyor's situation — she can now calculate the area of any triangle within the shamba where she has measured two boundary sides and the angle between them, bringing her one step closer to finding the total area of the five-sided plot.

LESSON 4 (80 minutes (double lesson)): Area of a Triangle Using Heron's Formula	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to calculate the area of a triangle using Heron's formula when only the three side lengths are known, and to compare its results with the sine formula.
Knowledge	Learners will know that Heron's formula $A = \sqrt{[s(s-a)(s-b)(s-c)]}$, where $s = ½(a+b+c)$ is the semi-perimeter, can calculate the area of any triangle given only its three side lengths, without needing a height or angle measurement.
Skills	Learners will be able to: (a) compute the semi-perimeter s of a triangle given three sides; (b) apply Heron's formula step-by-step to find the area of a triangle; (c) verify results using the sine formula where sufficient information is available; (d) decompose an irregular polygon into triangles and apply Heron's formula to each component triangle.
Attitudes	Learners will appreciate that multiple mathematical tools exist for the same problem, and that choosing the right formula depends on the information available — a critical real-world decision-making skill for land surveyors, carpenters, and engineers in Kenya.
Key Inquiry Question	How do we determine the area of a polygon? How do we apply area of polygons in real-life situations?
Purpose in Storyline	The Nakuru land surveyor has previously used ½ × base × height and the sine formula. Now, at a triangular sub-section of the shamba, neither a perpendicular height nor any angle is marked on the old survey map — only the three fence-post distances are recorded. A Kisumu carpenter faces the same problem: cutting a triangular wooden panel where only the three edge lengths are known. Learners must discover and apply Heron's formula to unlock the area from side lengths alone, directly advancing the driving question: which formula do I use when I don't know the height?
Safety Notes	No physical hazards in this lesson. Ensure calculators are handled responsibly. If using printed survey maps, take care with sharp rulers. Remind learners to double-check calculator mode (ensure it is NOT in radian mode when using the sine formula for verification).

B. LESSON OVERVIEW
In Lesson 4 of 8, learners encounter a new constraint in the Nakuru shamba storyline: a triangular sub-plot of the farm (and a parallel scenario of a Kisumu carpenter's triangular wooden panel) for which only the three side lengths are known — no height, no angle. Learners first predict whether area can be found without height or angle, then observe a guided derivation of Heron's formula via the worked relationship between the cosine rule and the sine formula. Learners apply the formula to at least three graded Kenyan-context problems, verify one result using the sine formula (cross-checking tools), and discuss which formula is most powerful and in what situations. The lesson closes with learners updating the class Driving Question Board and revising their cumulative polygon-decomposition model of the Nakuru shamba to include Heron's formula as a new tool in their surveyor's toolkit.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Parallelogram on]()	Learners are presented with a printed sketch (or board drawing) of the Kisumu carpenter's	Display the triangular panel sketch on the board. Read aloud: 'A carpenter in Kisumu is cutting a	Predict-Before-Instruction (Anticipatory Activation): By committing a prediction to paper	Observable indicator: Can the learner articulate specifically WHY the known formulas fail (e.g., 'I

the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Quadratic Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	triangular wooden panel labelled: side a = 7 cm, side b = 9 cm, side c = 11 cm. No angles, no height markings. In their exercise books, learners individually write responses to two prompts: (1) 'Can you calculate the area of this triangle? Write YES or NO and explain your reasoning in one sentence.' (2) 'If YES, write the formula or method you would use. If NO, write what extra information you would need.' After 5 minutes of individual thinking, learners pair-share with a neighbour for 2 minutes. Selected pairs share with the class. Learners record predictions in their ARES journals under the heading: 'My Prediction — Lesson 4.'	triangular wooden panel for a door frame. The three side lengths are 7 cm, 9 cm, and 11 cm. No angles are marked. No height is drawn. Can you find the area?' Allow 5 minutes of SILENT individual writing — do not answer questions during this time. After 5 minutes, say: 'Turn to your neighbour. You have 2 minutes — share your prediction and your reasoning.' After pair-share, cold-call EXACTLY 4 pairs using name cards: 2 pairs who said YES, 2 who said NO. For each, probe with: 'What formula are you thinking of, and what stops you from using it right now?' Do NOT confirm or deny correctness. Record YES/NO tally on the board. Say: 'By the end of today, every one of you will be able to solve this — and you will understand WHY a new formula is needed.'	before instruction, learners activate prior knowledge of ½bh and the sine formula, surface their own gaps (no height, no angle), and create cognitive dissonance that motivates the new formula. The YES/NO tally makes the class's collective uncertainty visible and sets up the 'need to know.'	need a height' or 'I need at least one angle for the sine formula')? Teacher scans written responses during pair-share — look for learners who write only 'I don't know' (flag for targeted support during Phase 3). Learners who correctly identify the constraint show readiness for the derivation phase.
Observe Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Quadratic Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners follow a structured guided-derivation activity in their ARES workbooks. The workbook provides a step-by-step scaffold showing how Heron's formula emerges from combining the cosine rule (to find an angle) with the sine area formula — presented as a sequence of numbered steps with blank boxes for learners to fill in key values. Learners work through the scaffold individually, filling in blanks as the teacher narrates each step at the board. Learners then read a short illustrated 'Surveyor's Field Note' insert in their workbook: a 1-paragraph story of the Nakuru surveyor dividing the shamba's triangular northern sub-plot (sides 40 m, 55 m, 63 m) and identifying that only side lengths appear on the old boundary map. The insert	Say: 'We are going to watch a formula being BORN — not just memorised. Follow every step in your workbook on page __.' Work through the derivation on the board in clearly numbered steps: STEP 1 — Write the cosine rule: $\cos C = (a^2+b^2-c^2)/(2ab)$. STEP 2 — Express sin C from cos C using $\sin^2 C + \cos^2 C = 1$. STEP 3 — Substitute into $A = ½ab \sin C$. STEP 4 — Simplify using the factorisation of difference of squares. STEP 5 — Introduce $s = (a+b+c)/2$ and show that the expression under the root becomes $s(s-a)(s-b)(s-c)$. At each step, pause for WAIT TIME of 12 seconds before filling in the next line. At Step 4, cold-call 2 learners to suggest the next simplification. After completing the derivation, say: 'Box that formula.	Guided Derivation with Structured Scaffold (Making Thinking Visible): Rather than presenting Heron's formula as a 'given,' the step-by-step derivation helps learners understand its logical origin, making the formula memorable and meaningful. The blank-box scaffold ensures all learners are cognitively engaged at every step rather than passively copying. Connecting the derivation to the Nakuru shamba storyline anchors the abstract algebra to a real need.	Observable indicator: Learners correctly fill in the five scaffold boxes without prompting by Step 5. Teacher circulates and checks that (a) the semi-perimeter definition is written correctly (s = ½ perimeter, NOT ½ area), and (b) the formula is copied with the square root sign enclosing the ENTIRE product. Flag any learner who writes s as the area — this is a critical misconception to address in Phase 3.

	ends with: 'The surveyor recalled a formula from school — one that needs only three sides...' Learners copy the final boxed formula into their journals: A = √[s(s−a)(s−b)(s−c)], where s = (a+b+c)/2.	Write it in RED. You will use it for the rest of your life as a surveyor, carpenter, or engineer.' Then narrate the Surveyor's Field Note aloud while learners follow in their workbooks. End by asking: 'Look at the formula — what is s? Turn to page __ and fill in the definition box.'		
Explain Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Quadratic Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners work through three graded application problems in their ARES workbooks, all set in Kenyan contexts. Problem 1 (Guided, whole-class): The Kisumu carpenter's triangular panel — sides 7 cm, 9 cm, 11 cm. Learners solve step-by-step alongside the teacher. Problem 2 (Pair work): The northern triangular sub-plot of the Nakuru shamba — sides 40 m, 55 m, 63 m. Pairs must (a) calculate area using Heron's formula, and (b) verify by first using the cosine rule to find one angle, then applying the sine formula A = ½ab sin C. They record both answers and check if they match. Problem 3 (Individual challenge): A triangular maize farm near Kericho has sides 120 m, 85 m, and 97 m. The farmer wants to know the area to calculate how many bags of fertiliser to buy (one bag covers 500 m²). How many bags does the farmer need? Learners who finish early tackle an extension: 'Which formula — sine or Heron's — would you recommend to the Nakuru surveyor as most powerful, and why? Write 3 sentences.' Groups share Problem 2 verification findings with the class in the final 5 minutes of this phase.	For Problem 1, narrate each step aloud: 'STEP 1 — write down a, b, c. STEP 2 — calculate s. s = (7+9+11)/2. What is that? Cold-call ONE learner.' Wait 10 seconds. 'STEP 3 — calculate s−a, s−b, s−c. Write all three. STEP 4 — multiply s(s−a)(s−b)(s−c). STEP 5 — take the square root. Round to 2 decimal places.' Write the answer: A ≈ 29.39 cm². Say: 'Now check your prediction from Phase 1. Were you right that this was possible?' For Problem 2, say: 'With your partner — solve it TWICE. Once with Heron's, once with cosine rule + sine formula. Do your two answers agree? If they don't, find your error before I come to you.' Circulate — spend 30 seconds per pair, checking semi-perimeter calculation first. After 8 minutes, cold-call 2 pairs to share their verification step. For Problem 3, say: 'This is now real surveying. Show ALL working — a Kericho farmer is depending on your calculation.' After 5 minutes, cold-call 1 learner for the final bag count answer and ask: 'Did you round UP or DOWN, and why?' (Expected answer: always round UP — you cannot buy half a bag.)	Graduated Release with Cross-Formula Verification (Productive Struggle + Self-Checking): The three-problem sequence moves from guided to paired to independent, building confidence incrementally. The verification task in Problem 2 — deliberately using TWO formulas to check the same answer — develops a mathematician's habit of checking work and deepens understanding that Heron's formula and the sine formula are consistent tools, not competing ones. The real-world application in Problem 3 (fertiliser bags) connects mathematics to agricultural decision-making, grounding abstract computation in Kenyan rural life.	Observable indicators: (1) In Problem 1, all learners produce s = 13.5 correctly — if not, the semi-perimeter definition was not understood; re-teach immediately. (2) In Problem 2, both methods must yield the same area (≈ 1,090.5 m²) — mismatches reveal arithmetic errors or formula confusion. (3) In Problem 3, the correct area is ≈ 4,028.6 m², requiring 9 bags (rounding up from 8.06) — learners who write 8 bags have not applied the real-world rounding reasoning. Teacher uses a show-of-hands check: 'Raise your hand if your two answers in Problem 2 matched exactly.' Count and record proportion for lesson reflection.
Driving Question	Learners return to the class Driving Question Board (DQB)	Distribute sticky notes or index cards. Say: 'You have 4 minutes	Driving Question Board Update (Public Knowledge Building): The	Observable indicator: Evidence cards should reference Heron's

Board (DQB) Creation 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Quadratic Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	displayed at the front of the room. In pairs, learners write on a sticky note or card: (1) ONE piece of evidence gathered today that helps answer the driving question — 'How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot, no matter what measurements are available?' and (2) ONE new question that has emerged. Pairs post their evidence card in the 'EVIDENCE' column and their question card in the 'STILL WONDERING' column of the DQB. The class briefly reviews the board — teacher reads 3 evidence cards aloud and asks: 'How does today's formula change what the Nakuru surveyor can do that they couldn't do in Lesson 3?' Learners also update their personal 'Formula Toolkit' tracker (a running table in their ARES journal listing: formula name, what you need to know, what you can find).	— write ONE thing you learned today that the Nakuru surveyor can now use, and ONE question still in your mind.' After posting, stand at the DQB and read 3 evidence cards aloud (choose one strong, one partial, one that reveals a misconception). For the misconception card, do NOT identify the author — instead say: 'Someone wrote this — what do we think? Is this exactly right, or can we improve it?' Allow 3 responses. Then ask the connecting question: 'Last lesson the surveyor needed an angle. Today, what does the surveyor need? Has their toolkit gotten stronger or weaker?' Wait 12 seconds. Cold-call 2 learners. End by pointing at the DQB: 'The board is filling up. Every lesson, the surveyor gets more powerful. By Lesson 8, we will answer the full driving question.'	DQB makes the class's growing understanding visible and cumulative. By requiring learners to articulate EVIDENCE (not just 'what I learned'), the strategy develops scientific and mathematical argumentation habits. The 'Still Wondering' column normalises uncertainty and surfaces misconceptions for the teacher to address in subsequent lessons.	formula and the condition 'when only three sides are known' — vague cards ('I learned a new formula') indicate surface-level processing. Question cards that reference the broader phenomenon ('Can we use this on the five-sided shamba?') indicate deep engagement with the driving question. Teacher photographs the updated DQB for planning Lesson 5.
Model Building Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Quadratic Formula (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their cumulative 'Surveyor's Decomposition Model' — a diagram of the five-sided Nakuru shamba that they have been building across the lesson sequence. In earlier lessons, they labelled sub-triangles that could be solved using ½bh (Lesson 2) and the sine formula (Lesson 3). Today, they identify which sub-triangle(s) of the shamba could ONLY be solved using Heron's formula (i.e., where only three side lengths are available and no angle is marked). Learners annotate those triangles with a green label: 'Heron's formula — 3 sides known.' They also update a legend/key on their model diagram to include: 'Heron's Formula: A = √[s(s−a)(s−b)(s−c)].' In the final 2	Say: 'Open your shamba model from Lesson 2. You have been building this all unit. Today, you are adding a new tool.' Display a class version of the shamba model on the board. Point to the northern triangular sub-plot: 'This triangle — the surveyor ONLY has the three fence-post distances. No angle is recorded. Which lesson's formula fits here?' Wait 10 seconds. Cold-call 1 learner. Confirm: 'Heron's formula. Label it green.' Give learners 5 minutes to annotate their personal models and update their legend. Circulate and check that the formula is written correctly in the legend. In the final 2 minutes, ask: 'In one sentence — complete this: The Nakuru surveyor can now calculate the	Cumulative Model Revision (Iterative Modelling): The shamba decomposition diagram serves as a running visual argument that learners refine each lesson. Annotating specific triangles with specific formulas forces learners to make explicit decisions about which tool applies under which conditions — the core intellectual challenge of the unit. The model becomes a physical record of the class's growing capability to answer the driving question.	Observable indicator: Learners correctly identify at least one triangle in the shamba model where ONLY side lengths are available and label it with Heron's formula. The written sentence at the bottom of the model should mention 'three side lengths' and 'semi-perimeter' (or 's') as the key condition. Sentences that only say 'because we used Heron's formula' without specifying the condition are flagged for follow-up at the start of Lesson 5.

	minutes, learners write one sentence below their model: 'The Nakuru surveyor can now calculate the area of the northern triangular sub-plot because...'	area of the northern sub-plot because...' Cold-call 2 learners to read their sentence aloud. Affirm strong responses. Close with: 'Next lesson, we will zoom out from triangles to quadrilaterals — parallelograms, trapeziums, and kites. The shamba has more than just triangles.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record notes in your ARES teacher journal:

1. PREDICT PHASE — What proportion of learners said YES (area is possible with only three sides)? What prior knowledge were they drawing on, and was it accurate? Did the YES/NO tally reveal a class pattern worth addressing at the start of Lesson 5?

2. DERIVATION QUALITY — Did learners engage with the step-by-step scaffold, or did they copy passively? Which step caused the most hesitation? (Common bottleneck: Step 4 — the difference-of-squares factorisation. If more than 30% of learners stalled here, plan a 5-minute recap at the start of Lesson 5.)

3. VERIFICATION TASK — In Problem 2, what proportion of pairs successfully verified that both formulas give the same area? Mismatches almost always arise from: (a) wrong angle computed from the cosine rule, or (b) calculator in radian mode. Note how many pairs needed intervention and what the root cause was.

4. REAL-WORLD REASONING — In Problem 3, did learners spontaneously round UP to 9 bags, or did most write 8? The rounding-up insight is a key numeracy habit. If fewer than half rounded correctly, build a brief 'real-world rounding' discussion into Lesson 5's opening.

5. DQB QUALITY — Were evidence cards specific (mentioning 'three sides,' 'semi-perimeter,' 'Heron's formula') or generic ('I learned a new formula')? Generic cards suggest the driving question connection was not made explicit enough. In Lesson 5, open by reading two contrasting evidence cards and asking the class to improve the weaker one.

6. MODEL BUILDING — Did learners correctly identify which sub-triangle of the shamba requires Heron's formula (rather than the sine formula)? If learners labelled ALL triangles with Heron's formula, they may not yet appreciate that it is a special-case tool for 'three sides only' — address this at the start of Lesson 5 before introducing quadrilaterals.

7. PACING — Was 80 minutes sufficient? The derivation (Phase 2) and the three problems (Phase 3) are the highest-risk phases for time overrun. If Phase 3 Problem 3 was not completed in class, set it as the opening activity for Lesson 5 rather than as homework, to preserve the fertiliser-bag real-world reasoning discussion.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed a step-by-step derivation showing how Heron's formula $A = \sqrt{s(s-a)(s-b)(s-c)}$ emerges algebraically from the cosine rule combined with the sine area formula. They also observed that the Nakuru surveyor's northern triangular sub-plot (sides 40 m, 55 m, 63 m) and the Kisumu carpenter's triangular panel (sides 7 cm, 9 cm, 11 cm) both lack height and angle data — making previously learned formulas insufficient without Heron's formula.
What did I learn?	Learners learned that: (1) the semi-perimeter $s = (a+b+c)/2$ is the essential first step in Heron's formula; (2) Heron's formula calculates the exact area of any triangle given only its three side lengths; (3) Heron's formula and the sine formula are consistent — they yield the same area when both are applicable, which allows cross-checking; (4) real-world applications (e.g., calculating fertiliser bags for a Kericho maize farm) require rounding area-based answers UP to ensure sufficiency.
How does this explain the	Learners explained that the Nakuru land surveyor can now calculate the area of any triangular sub-plot of the shamba even when

phenomenon?	only fence-post distances (side lengths) are recorded and no angles or heights are available, by applying Heron's formula. They updated their cumulative shamba decomposition model to label the northern triangular sub-plot as a 'Heron's formula triangle,' and articulated in writing the specific condition that makes Heron's formula the appropriate tool: three side lengths known, no height or angle available.

LESSON 5 (80 minutes): Polygons: Classification, Properties, and Decomposition into Triangles	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to classify polygons and recognise that any polygon can be decomposed into triangles, connecting this insight to the calculation of area of irregular polygons such as the five-sided shamba plot in Nakuru.
Knowledge	By the end of the lesson, the learner should be able to: (a) classify polygons as regular or irregular, and as convex or concave in different situations; (b) identify properties of common polygons including the number of sides, vertices, and interior angles; (c) decompose any polygon into non-overlapping triangles and explain why this is a valid strategy for area calculation.
Skills	By the end of the lesson, the learner should be able to: (a) sort and classify cut-out polygon shapes using observable properties; (b) draw diagonal lines to decompose polygons into triangles; (c) connect polygon decomposition to the area formulas already learned (sine rule and Heron's formula) to plan a solution for an irregular polygon.
Attitudes	The learner appreciates the logical structure of mathematics and develops confidence in approaching complex, real-world problems by breaking them into manageable parts; the learner collaborates respectfully with peers during classification activities.
Key Inquiry Question	How do we determine the area of a polygon? How do we apply area of polygons in real-life situations?
Purpose in Storyline	In Lessons 1–4, learners built tools: the sine area formula and Heron's formula for triangles, and area formulas for parallelograms, trapeziums, and kites. Now, in Lesson 5, learners turn their attention to the five-sided shamba plot itself. Before they can calculate its area, they must understand what kind of shape it is and discover the powerful insight that ANY polygon — no matter how irregular — can be cut into triangles. This is the bridge that will allow them to solve the anchoring phenomenon in Lessons 6–8.
Safety Notes	Scissors are used during the cut-out activity. Remind learners to handle scissors carefully, pass them handle-first, and keep them flat on the desk when not in use. No other safety concerns anticipated.

B. LESSON OVERVIEW

Learners begin by revisiting the Nakuru surveyor's five-sided shamba plot and asking: what kind of polygon is it, and does its classification affect how we find its area? Through a hands-on cut-out sorting activity, learners classify a set of polygon cards as regular/irregular and convex/concave, anchoring definitions in observable properties. A short video resource deepens understanding of interior angle sums and polygon naming. The pivotal sensemaking moment comes when learners physically cut a paper pentagon into triangles and count how many non-overlapping triangles fill it — discovering the rule $(n - 2)$ triangles for an n-sided polygon. Learners then map this insight directly onto the surveyor's shamba plot, sketching the decomposition and identifying which triangle-area formula (sine or Heron's) they would apply to each triangle given the available measurements. The lesson closes with an updated Driving Question Board entry and a cumulative model revision connecting all five lessons.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown a sketch of the surveyor's five-sided shamba	Draw the irregular pentagon on the board clearly, labelling vertices A–	Prediction with Productive Tension: Learners commit to a	Observable indicator: Every learner has a written prediction

📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Polygon Classification in the Coordinate Plane (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	plot (drawn on the board and in their workbooks) with its five fence-post corners labelled A, B, C, D, E and some side lengths written in metres. The teacher poses the prompt: 'Without doing any calculation, predict — how many triangles do you think you would need to completely fill this five-sided plot with no gaps and no overlaps? Write your prediction and one reason in your exercise book.' Learners write individually for 2 minutes, then share with a shoulder partner for 1 minute.	E. Say: 'This is the shamba in Nakuru. Today we are NOT calculating the area yet — we are going to understand the shape first.' Pose the prediction prompt verbally and write it on the board: 'How many triangles fill this five-sided plot?' Allow 2 minutes of silent individual thinking. Circulate and scan — do NOT correct predictions. After 2 minutes say: 'Share your prediction and reason with your partner. You have 1 minute.' Cold-call 3 learners with varied predictions. For each, ask: 'What made you choose that number?' Use WAIT TIME of 12 seconds after each cold-call before taking the answer. Record the three predictions on the board (e.g. 3, 4, 5) without indicating which is correct — leave productive tension. Say: 'By the end of today, you will know the answer and, more importantly, WHY.'	specific numerical prediction before instruction. Recording multiple conflicting predictions on the board creates cognitive dissonance that motivates engagement with the evidence-gathering activity. This activates prior spatial reasoning about triangles built in Lessons 1–4.	AND a written reason in their exercise book within 2 minutes. Teacher scans books during circulation — if more than one-third of learners have left the reason blank, pause and prompt: 'Think about how many sides a triangle has compared to a pentagon.' Listen to partner-share conversations for misconceptions such as 'you need 5 triangles because there are 5 sides' — note these for targeted address in Phase 3.
Observe Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Polygon Classification in the Coordinate Plane (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Activity A — Polygon Sort (20 minutes): Each group of four receives an envelope containing 10 pre-cut cardboard polygon shapes (triangles, quadrilaterals, pentagons, hexagons — some regular, some irregular, some convex, some concave — labelled P1–P10). Groups also receive a sorting mat with four quadrants: Regular/Convex, Regular/Concave, Irregular/Convex, Irregular/Concave. Learners physically place each shape on the mat, discuss as a group, and record their placement decisions and reasoning in a table in their workbooks. Activity B — Video Resource (10 minutes): Learners watch a short curated video (or	Distribute envelopes and sorting mats. Say: 'In your group, sort these shapes into the four quadrants. You must agree as a group AND write one reason for each placement.' Allow 8 minutes. Circulate — when a group is uncertain, ask: 'Does this shape have all equal sides and all equal angles? If yes, it is regular. If you poke a vertex inward, does the shape cave in? If yes, it is concave.' Do NOT give answers — ask guiding questions only. After 8 minutes, cold-call 2 groups to share placements for shapes P3 (an irregular convex pentagon — the shamba shape) and P7 (a concave hexagon). Use WAIT TIME of 10 seconds after asking each group to respond. Facilitate a	Concrete-to-Abstract Bridging via Manipulatives: Physical sorting of cut-out shapes gives learners direct sensory experience of polygon properties before formal definitions are introduced. The pentagon cut-and-fill activity makes the decomposition strategy tangible — learners literally see the triangles inside the polygon. This grounds the abstract formula $(n - 2)$ in physical experience, which is especially important for visual and kinaesthetic learners.	Observable indicators: (1) Sorting mat — check that each group has correctly identified P3 as irregular and convex (the shamba-type shape). If groups place it in the wrong quadrant, address during the cold-call debrief. (2) Guided note-frame — learners should correctly write $S = (n - 2) \times 180°$ and compute $S = 540°$ for the pentagon. Scan books during circulation. (3) Cut-and-fill count — learners should arrive at 3 triangles for the pentagon and 4 for the hexagon. If a learner reports a different number, check whether they drew diagonals from a single vertex only. Ask: 'Did all your diagonals start from the same corner?'

	teacher-led slide presentation if video is unavailable) that covers: polygon naming by number of sides, the interior angle sum formula S = (n − 2) × 180°, and examples of convex vs concave polygons. Learners complete a guided note-frame in their workbooks as they watch. Activity C — Pentagon Cut-and-Fill (8 minutes): Each learner receives a paper copy of the same irregular pentagon as the shamba plot. Learners use a ruler to draw straight diagonal lines from ONE vertex to all non-adjacent vertices, cutting the pentagon into triangles. They count the triangles, record the number, and repeat with a hexagon cut-out.	brief whole-class discussion: 'Did any groups disagree? Why?' Transition to video: 'Now let us watch/review to check our sorting and learn the interior angle sum rule.' Pause the video or slides at the interior angle sum formula. Ask: 'For our five-sided shamba plot, what is the sum of all interior angles? Work it out now.' WAIT TIME 12 seconds, then cold-call 1 learner. For the cut-and-fill activity, model the first diagonal on the board pentagon before learners begin. Say: 'Draw from vertex A only — go to every vertex that is NOT next to A. Count your triangles.'		
Explain Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Polygon Classification in the Coordinate Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners reconvene as a whole class for structured sensemaking. The teacher leads a guided discussion to formalise: (1) the definition and examples of regular vs irregular and convex vs concave polygons; (2) the interior angle sum formula S = (n − 2) × 180°; (3) the triangle decomposition rule: an n-sided polygon decomposes into (n − 2) triangles drawn from one vertex. Learners then apply this directly to the Nakuru shamba: they sketch the five-sided plot in their books, draw the two diagonals from vertex A to create 3 triangles, and label each triangle (Triangle 1: ABC, Triangle 2: ACD, Triangle 3: ADE). They then identify — for each triangle — what information the surveyor has (sides, angles) and which area formula (sine rule: A = ½ab sin C, or Heron's formula: A = √[s(s−a)(s−b)(s−c)]) would be most appropriate to use. This is a planning exercise, not yet a	Return to the board predictions from Phase 1. Say: 'Raise your hand if you predicted 3 triangles.' Pause. 'Let us now prove why 3 is correct for any pentagon.' Write on the board: 'Number of triangles = n − 2. For n = 5: 5 − 2 = 3.' Ask: 'Why does this rule work? Discuss with your partner for 90 seconds.' Cold-call 2 pairs — WAIT TIME 12 seconds each. Guide learners toward the idea that each diagonal 'uses up' one side-worth of the polygon. Formalise definitions on the board with examples drawn from the sorting activity shapes. Then say: 'Open to your shamba sketch. Draw the two diagonals from vertex A now.' Circulate — 2 minutes. Then ask: 'You have three triangles. For Triangle ABC, the surveyor knows sides AB, BC, and angle B. Which formula do we use?' WAIT TIME 15 seconds. Cold-call 1 learner. Say: 'Say the formula aloud for the class.' Repeat for Triangle ACD (where	Worked Example with Live Annotation: The teacher annotates the board pentagon in real time as learners annotate their own sketches simultaneously. This dual-coding approach (visual diagram + symbolic formula labels) helps learners connect the geometric decomposition to the algebraic tools they already have. The explicit formula-selection step (sine vs Heron's) consolidates Lessons 1–4 learning and makes the pathway to solving the phenomenon concrete and achievable.	Observable indicators: (1) All learners should have a correctly decomposed pentagon sketch with 3 labelled triangles in their books. Scan during the 2-minute drawing period. (2) During cold-calls on formula selection, listen for correct justification: 'I use the sine formula when I have two sides and the included angle' and 'I use Heron's formula when I have all three sides but no height.' If a learner selects the wrong formula, ask: 'What information does the surveyor have for that triangle? What does the sine formula need?' (3) Exit check: after writing the total area equation on the board, ask learners to copy it and annotate each term with the formula they would use. Collect and check 5 books at random.

	calculation.	only three sides are known → Heron's formula) and Triangle ADE (where two sides and included angle are known → sine formula). Write on the board: 'Total Area of Shamba = Area(ABC) + Area(ACD) + Area(ADE).' Say: 'This is the plan. In Lessons 6 and 7, we will execute this plan with real numbers.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Polygon Classification in the Coordinate Plane (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner writes a new sticky note (or entry in the designated DQB section of their exercise book) responding to the prompt: 'What new understanding from today moves us closer to answering: How can the Nakuru land surveyor calculate the exact area of the five-sided shamba plot?' Learners also review the class DQB from Lessons 1–4 and identify one question that today's lesson has now answered and one new question that today's lesson has raised. Groups share their 'answered' and 'new' questions briefly (2 minutes total), and the teacher updates the physical or digital DQB accordingly.	Display the full Driving Question on the board: 'How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot, no matter what measurements are available?' Say: 'On your sticky note, write one sentence: what does today's lesson give the surveyor that they did not have before?' Allow 2 minutes of silent writing. Then say: 'Look at our DQB from the last four lessons. Find one question that you can now answer. Underline it.' Cold-call 2 learners to share their answered question and their new question. WAIT TIME 10 seconds after each. Add the new questions to the DQB. Say: 'Notice that our board is filling up with evidence. Each lesson is a tool. Today we gave the surveyor a plan — decompose the plot into triangles.'	Driving Question Board as Cumulative Evidence Tracker: The DQB makes visible the growing body of knowledge across the lesson sequence. By identifying both answered and newly raised questions, learners experience science/mathematics as an iterative inquiry process, building metacognitive awareness of what they know and what remains to be solved.	Observable indicator: Each learner's DQB entry should explicitly mention polygon decomposition or 'cutting into triangles' as the new understanding. Entries that only restate definitions (e.g. 'I learned what a convex polygon is') indicate surface-level engagement — follow up with: 'How does knowing what kind of polygon the shamba is help the surveyor calculate area?'
Model Building Phase 📹 **VIDEO:** [Parallelogram on the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:**	Learners return to their cumulative 'Surveyor's Toolkit' model — a diagram begun in Lesson 1 that grows each lesson. In Lesson 5, learners add a new layer: they sketch the five-sided shamba polygon, draw the decomposition diagonals, label the three triangles, and annotate each triangle with the formula to be used and the information available. They also add a summary statement at the bottom of their model: 'Any	Say: 'Open to your Surveyor's Toolkit model. You are adding Lesson 5 today. Your model should now show: the three triangle formulas from Lessons 1–4 AND the decomposition plan for the five-sided shamba.' Circulate for 5 minutes, giving specific feedback. Look for: correct labelling of triangles by their vertices, correct annotation of which formula applies to each, and the summary statement written in	Cumulative Model Revision (Toolkit Building): The Surveyor's Toolkit is a living document that grows across all 8 lessons, making learning progression visible and concrete. Adding the decomposition layer in Lesson 5 integrates all prior formula knowledge into a unified problem-solving strategy, reinforcing that mathematics builds cumulatively toward solving complex real-world problems.	Observable indicator: Learner models should contain all five lessons' contributions by the end of Lesson 5 — (1) sine area formula for triangles, (2) Heron's formula, (3) quadrilateral area formulas, (4) regular polygon formula, and now (5) polygon decomposition plan with labelled triangles for the shamba. If a learner's model is missing earlier entries, use the pair-comparison step to prompt them to fill gaps.

Polygon Classification in the Coordinate Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	polygon with n sides can be decomposed into (n − 2) triangles. The total area = sum of areas of all triangles.' Learners who finish early extend by decomposing a hexagonal plot sketch (given as an extension problem) and writing the total area plan for 4 triangles.	the learner's own words. If a learner's model is incomplete from previous lessons, direct them to the class reference poster. With 3 minutes remaining, say: 'In pairs, compare your Toolkit models. Find one thing your partner has that you are missing and add it now.' Close by saying: 'Next lesson, we will use this exact plan with real measurements from the shamba to calculate the actual area. Your model is your roadmap.'		Collect 4–5 Toolkit models at the end of class to review depth of annotation and return with written feedback at the start of Lesson 6.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the cut-and-fill activity, did all learners successfully arrive at 3 triangles for the pentagon? If several learners drew diagonals from multiple vertices (getting more than 3 triangles), plan a 5-minute re-teach at the start of Lesson 6 using a large paper model on the board. (2) During the Explain phase, could learners correctly select between the sine formula and Heron's formula for each triangle in the shamba decomposition, or did they confuse the two? If formula selection was uncertain, prepare a one-page reference card with the decision rule: 'Two sides + included angle → sine formula. Three sides, no height → Heron's formula.' (3) Was the DQB update substantive — did learners connect decomposition to area calculation — or did entries remain at the definitional level? Consider adding a sentence starter on the board in Lesson 6: 'Because the shamba can be cut into triangles, the surveyor can now…' (4) Were the cut-out polygon envelopes prepared and distributed efficiently, or did setup time reduce learning time? If so, pre-assign a group materials monitor for Lesson 6. (5) Do learners' Toolkit models show genuine integration of all five lessons' tools, or are entries from Lessons 1–4 missing or incomplete? Use collected models to identify the 3–4 learners who need targeted support before Lesson 6's calculations begin.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners sorted 10 pre-cut polygon shapes (regular/irregular, convex/concave) onto a four-quadrant sorting mat and physically cut a paper irregular pentagon into triangles by drawing diagonals from a single vertex, counting 3 triangles. They repeated this with a hexagon, finding 4 triangles.
What did I learn?	Any polygon with n sides can be decomposed into (n − 2) non-overlapping triangles by drawing diagonals from one vertex. For the five-sided shamba plot in Nakuru, this gives 3 triangles. The total area of the polygon equals the sum of the areas of these triangles, each calculated using either the sine area formula or Heron's formula depending on the available measurements.
How does this explain the phenomenon?	The surveyor cannot use a single formula directly on the five-sided irregular shamba plot because it is not a standard shape. By classifying it as an irregular convex pentagon and decomposing it into 3 triangles (ABC, ACD, ADE), the surveyor can apply the triangle area tools from Lessons 1–4 to each triangle separately and add the results. This decomposition strategy works for ANY polygon, making it a universal tool for land area problems.

LESSON 6 (80 minutes (2 lessons of 40 minutes each)): Area of Regular Polygons: Central Triangle Decomposition	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To derive and apply formulae for the area of regular polygons using the central triangle decomposition method.
Knowledge	By the end of the lesson, the learner should be able to: (a) decompose regular polygons into congruent central triangles; (b) derive the general formula $A = (n \times s^2) / (4 \times \tan(180°/n))$ for the area of a regular polygon; (c) calculate the area of regular polygons including equilateral triangles, squares, regular pentagons, regular hexagons, and regular octagons.
Skills	By the end of the lesson, the learner should be able to: (a) apply the central triangle decomposition strategy to find the area of any regular polygon; (b) use a calculator or mathematical tables to evaluate $\tan(180°/n)$ for various values of n; (c) apply the regular polygon area formula to real-life tiling and land-surveying contexts.
Attitudes	The learner should: (a) appreciate the elegance and efficiency of a single formula that replaces multiple specific formulae; (b) show curiosity in connecting geometric decomposition to practical surveying and design problems; (c) value precision and accuracy when computing areas for real-world applications such as land allocation and architectural tiling.
Key Inquiry Question	How do we determine the area of a polygon? How do we apply area of polygons in real-life situations?
Purpose in Storyline	In Lesson 6, learners step back from the irregular shamba and examine a special class of polygon — the regular polygon — whose perfect symmetry allows a single, elegant formula. By deriving $A = (n \times s^2) / (4 \times \tan(180°/n))$ through central triangle decomposition and applying it to the hexagonal tile pattern of Nairobi's Jamia Mosque floor, learners add a powerful new tool to the surveyor's toolkit. This prepares them for Lesson 7, where irregular polygon areas are found by combining decomposition strategies with all previously learned formulae.
Safety Notes	No physical hazards anticipated. Ensure learners handle geometry sets (compasses, rulers, protractors) carefully. Remind learners not to point compass tips toward themselves or peers.

B. LESSON OVERVIEW
Learners begin by revisiting the Nakuru surveyor's challenge and noting that the shamba's five sides are all unequal — but they ask: what if a plot WERE perfectly regular? This motivates exploration of regular polygons. Using ruler, compass, and protractor, learners construct a regular hexagon and a regular pentagon, then draw all radii from the centre to each vertex, revealing congruent isoceles triangles. They measure, calculate triangle areas, and generalise. Through guided inquiry, learners derive the formula $A = (n \times s^2) / (4 \times \tan(180°/n))$ and apply it to calculate the area of hexagonal ceramic floor tiles at Nairobi's Jamia Mosque — determining how many tiles are needed to cover a given floor section. The lesson closes with learners updating their Driving Question Board and revising their cumulative polygon decomposition model.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Parallelogram on]	Learners are shown a photograph of the hexagonal floor tile pattern at Nairobi's Jamia Mosque and a	Display the two images side by side on the board (or printed A3 sheets if no projector). Say: 'Look	Predict-Before-Instruction (Anticipation): Activates prior knowledge of area, surfaces	Observable indicator: Every learner has a written numerical prediction (even if incorrect) with a

the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Polygon Classification in the Coordinate Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	photograph of a honeycomb from a beehive in Kericho. The teacher poses: 'A single hexagonal tile has a side length of 20 cm. Without measuring every height or diagonal, predict: what is the area of one tile? Write your prediction in cm^2 in your exercise book. Also predict: how many such tiles would cover a rectangular section of floor that is 3 m × 2 m?' Learners write individual predictions and share with a shoulder partner, justifying their reasoning.	carefully at both images. What shape do you see repeated in both?' (WAIT 10 seconds, cold-call 3 learners.) Then say: 'The hexagon appears in nature and in human architecture — there must be a reason it is so useful. Today we will find out why, and we will find a formula that unlocks its area — and the area of ANY regular polygon.' Distribute the prediction prompt card. Circulate and note 2–3 interesting or divergent predictions to revisit in Phase 3. Do NOT correct predictions yet.	misconceptions about whether standard formulae apply to hexagons, and creates cognitive readiness for the derivation to follow.	brief written justification. Teacher scans books during circulation — flag learners who write 'I don't know' for targeted support in Phase 2.
Observe Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Polygon Classification in the Coordinate Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Activity A — Construct and Decompose (25 min): In groups of 4, learners use compass and ruler to construct (i) a regular hexagon with side 6 cm and (ii) a regular pentagon with side 6 cm on plain paper. They then draw straight lines from the centre of each polygon to every vertex, decomposing each polygon into congruent triangles. Learners label: n (number of sides), s (side length), the central angle at the apex of each triangle, and measure the perpendicular height (apothem) of one triangle using their ruler. They record all measurements in a structured observation table: | Polygon | n | s (cm) | Central angle | No. of triangles | Apothem h (cm) | Area of 1 triangle (cm^2) | Total area (cm^2) |. Activity B — Spot the Pattern (5 min): Learners compute: central angle = 360°/n for each polygon. They also compute tan(central angle / 2) and link it to the ratio (s/2) ÷ apothem. They record the relationship: apothem = (s/2) / tan(180°/n).	Before construction begins, say: 'The key question today is — can you find the area of one of these central triangles using only the side length s and the number of sides n? Let that guide your measurements.' (WAIT 10 seconds.) Circulate to each group during Activity A. At the 10-minute mark, pause the class: 'Hold up — how many triangles did your hexagon split into?' (Cold-call 3 groups — expected answer: 6.) 'And your pentagon?' (Expected: 5.) 'So for any regular polygon with n sides, how many triangles will you always get?' (WAIT 12 seconds, cold-call 2 learners — expected: n triangles.) During Activity B, prompt struggling groups: 'Look at one triangle only — you have the base s and you need the height. What trigonometric ratio connects the half-base, the height, and the angle at the top?' If a group finishes early, challenge them: 'Predict the apothem of a regular octagon with s = 6 cm before we compute it formally.'	Structured Observation with Data Table: Forces learners to record raw measurements before generalising, building evidence-based reasoning. The two-polygon comparison (hexagon vs pentagon) makes the pattern of central angle = 360°/n visible through contrast, not just assertion.	Observable indicator: Each group's observation table is fully completed with measured and calculated values. Teacher checks that the central angle column shows 60° for hexagon and 72° for pentagon — groups with wrong values are redirected to re-measure. At least one learner per group can state the relationship between apothem, s, and tan(180°/n) verbally before Phase 3 begins.
Explain Phase	Whole-class derivation (15 min):	At the start of derivation, say: 'We	Collaborative Algebraic Derivation	Observable indicator: During

📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Polygon Classification in the Coordinate Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher guides learners to derive the general formula step-by-step, with learners contributing each algebraic step. Starting from: Area of one central triangle = ½ × base × height = ½ × s × apothem. Substituting apothem = (s/2) / tan(180°/n) gives: Area of 1 triangle = ½ × s × (s/2) / tan(180°/n) = s^2 / (4 tan(180°/n)). Since there are n such triangles: Total area = n × s^2 / (4 tan(180°/n)) → A = (n × s^2) / (4 × tan(180°/n)). Learners copy the derivation into their notebooks, then immediately apply it: (i) Verify their measured hexagon area (n=6, s=6 cm) using the formula and compare with their table measurement — percentage error check. (ii) Calculate the area of one Jamia Mosque hexagonal tile with s = 20 cm. (iii) Calculate how many tiles are needed to cover a 3 m × 2 m floor section (accounting for no gaps in a regular hexagonal tiling). Paired practice (10 min): Learners solve two further problems: (a) A regular pentagonal flower bed at the Nairobi Arboretum has a side length of 3.5 m — find its area. (b) A boda-boda stage owner in Kisumu wants to pave a regular octagonal waiting bay with side 2 m — find the total paved area.	are going to build this formula together — I will not give you the steps, you will give them to me.' Write only 'A = ?' on the board. Ask: 'What is the area of ONE central triangle in terms of s and the apothem?' (WAIT 12 seconds, cold-call 1 learner.) After each step, pause and ask the class: 'Does everyone agree with this step? Thumbs up, thumbs sideways, or thumbs down.' Address any thumbs-down immediately. After the formula is derived, return to the predictions from Phase 1: 'You predicted the area of one hexagonal tile. Let's see how close you were — calculate now using the formula.' (WAIT 15 seconds for calculation.) Cold-call 2 learners to share their answers and their Phase 1 predictions — celebrate accuracy AND growth from incorrect predictions. For paired practice, say: 'You have 10 minutes. If you finish both problems, write one new word problem of your own using a regular polygon and a Kenyan context — we will solve two of them on the board.' Circulate and provide targeted prompts: 'What is n for an octagon?' and 'What does tan(22.5°) equal on your calculator?'	(Co-construction): Rather than receiving the formula, learners supply each algebraic step with teacher facilitation. This strategy builds ownership of the formula and deep understanding of why it works — not just how to apply it. Connecting back to the Phase 1 prediction creates a closing cognitive loop.	derivation, at least 4 different learners contribute a step unprompted. During paired practice, teacher checks that learners correctly substitute n and s — a common error is using the full central angle (360°/n) instead of the half-angle (180°/n) in the tangent. Any learner making this error is re-directed with: 'Look at your triangle — what is the angle at the APEX of ONE triangle, not the full rotation?'
Driving Question Board (DQB) Creation 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Each group receives two sticky notes of different colours. On the GREEN note, they write one new piece of evidence they now have to help answer: 'How can the Nakuru surveyor calculate the exact area of any irregularly shaped shamba?' On the ORANGE note, they write one remaining question or limitation: 'What can this formula NOT do for	Say: 'Our surveyor in Nakuru is watching us. She has just seen us derive a beautiful formula for regular polygons. Does this solve her problem completely?' (WAIT 10 seconds, cold-call 3 learners.) After responses, say: 'Write exactly what we have solved and exactly what remains unsolved for her.' After posting, read 2 green notes and 2 orange notes aloud	Driving Question Board (Evidence Tracking): The two-colour sticky note protocol forces learners to separately articulate evidence gained versus gaps remaining, making the boundary of their current knowledge visible and creating genuine motivation for Lesson 7.	Observable indicator: Every group's green note references the regular polygon formula with at least one correct variable (n or s). Every group's orange note correctly identifies that the surveyor's shamba is irregular and therefore the formula cannot be directly applied — groups that write vague statements ('it is hard') are prompted: 'What specific

for similar videos 📖 **HTML:** Polygon Classification in the Coordinate Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	the surveyor, and why?' Groups post their notes on the class DQB under the headings 'NEW EVIDENCE' and 'STILL WONDERING'. The class reads 3–4 posted notes aloud and the teacher annotates the board to highlight the emerging consensus: the regular polygon formula requires equal sides — it cannot be directly applied to the surveyor's irregular shamba. This sets up Lesson 7 (irregular polygons).	and ask: 'Do we all agree this is still a limitation? What kind of polygon is the surveyor's shamba — regular or irregular?' (Expected: irregular.) 'So what tool do we still need to build?' (Expected: a method for irregular polygons — previewing Lesson 7.) Update the DQB anchor strip: add 'Regular polygon formula (n sides): $A = (n \times s^2) / (4 \times \tan(180°/n))$' under 'FORMULAE WE KNOW'.		property of a regular polygon does the surveyor's shamba NOT have?'
Model Building Phase 📹 **VIDEO:** Parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Polygon Classification in the Coordinate Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their cumulative polygon decomposition model — a visual diagram they have been building since Lesson 1, showing the surveyor's shamba and the formulae tools available. In this lesson, they add a new 'branch' to the model: REGULAR POLYGONS → Central Triangle Decomposition → $A = (n \times s^2) / (4 \times \tan(180°/n))$, with labelled examples: hexagon (n=6), pentagon (n=5), octagon (n=8). They annotate with: 'Requires: ALL sides equal, ALL angles equal.' They also add a connecting arrow: 'If shamba were regular → use this formula. Since shamba is IRREGULAR → need Lesson 7 strategy.' Learners share their updated model with one other pair and give peer feedback using the sentence frame: 'Your model clearly shows ___. One thing to improve is ___.'	Say: 'Your model is the surveyor's growing toolkit. Every lesson, her toolkit gets more powerful. Today she gains the regular polygon tool — but she also gains clarity about its limits. Make sure your model shows BOTH.' Circulate during model building — look specifically for whether learners annotate the constraint ('requires equal sides'). After peer sharing, cold-call 1 pair to describe how their model has grown from Lesson 1 to Lesson 6: 'Walk us through your model — what did you add today and how does it connect to what came before?' Close with: 'Next lesson, we will give the surveyor the final tool she needs to crack the irregular shamba. Come ready.'	Cumulative Visual Model Revision (Progressive Model Building): By continually updating a single diagram across lessons, learners build a coherent, connected understanding of the entire unit rather than isolated formula memorisation. The explicit annotation of constraints (regular vs irregular) prevents overgeneralisation.	Observable indicator: The updated model clearly shows (a) the formula with correct notation, (b) at least two named polygon examples with values of n, and (c) an annotation distinguishing regular from irregular polygons. Teacher collects 4–5 models at random to review before Lesson 7 and provides written feedback on sticky notes returned at the start of the next lesson.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the derivation in Phase 3, were learners genuinely contributing algebraic steps, or were they passively copying? If the latter, plan a 'relay derivation' structure for Lesson 7 where each group derives one step and passes to the next. (2) Did learners correctly distinguish between the full central angle (360°/n) and the half-angle (180°/n) used in the tangent? Errors here indicate that the triangle decomposition diagram needs to be drawn

more explicitly — label the half-angle on the board diagram next time. (3) Were the Jamia Mosque and Kericho honeycomb contexts engaging for learners? Note which context generated more discussion and use it as the anchor in revision sessions. (4) Review the 4–5 collected cumulative models — do learners' constraint annotations ('requires equal sides') demonstrate understanding of why the formula cannot apply to the surveyor's irregular shamba? If not, begin Lesson 7 with a 5-minute whole-class comparison of a regular hexagon and the irregular shamba diagram, asking: 'What is the key difference?' (5) Which learners struggled with calculator use for tan(180°/n)? Plan a 3-minute targeted calculator drill at the start of Lesson 7 for those learners before the irregular polygon work begins.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners constructed regular hexagons and pentagons, drew radii from the centre to each vertex, and observed that every regular polygon decomposes into n congruent isoceles triangles. They measured the apothem and central angle of each triangle, and recorded data in a structured table — noting that the central angle always equals 360°/n and that the apothem relates to the side length via apothem = (s/2) / tan(180°/n).
What did I learn?	Learners derived the general formula for the area of any regular polygon: $A = (n \times s^2) / (4 \times \tan(180°/n))$, where n is the number of sides and s is the side length. They applied the formula to calculate the area of hexagonal floor tiles at Nairobi's Jamia Mosque (s = 20 cm), a pentagonal flower bed at the Nairobi Arboretum (s = 3.5 m), and a regular octagonal boda-boda waiting bay in Kisumu (s = 2 m).
How does this explain the phenomenon?	Learners explained that the regular polygon formula works because perfect symmetry allows every polygon to be split into identical central triangles, each with a calculable area using only the side length and the tangent of the half-central-angle. However, they also explained that this formula CANNOT be directly applied to the Nakuru surveyor's irregular shamba, because the shamba's sides are all unequal and its angles are not congruent — confirming the need for the irregular polygon decomposition strategy to be developed in Lesson 7.

LESSON 7 (80 minutes): Area of Irregular Polygons: Decomposing the Surveyor's Shamba	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To determine the area of irregular polygons by decomposing them into triangles and applying the sine rule, Heron's formula, and standard quadrilateral area formulas in real-life Kenyan land measurement contexts.
Knowledge	Students know that: (1) an irregular polygon can be decomposed into non-overlapping triangles; (2) the area of each triangle can be found using ½ab sin C (when two sides and an included angle are known), Heron's formula (when all three sides are known), or ½ × base × height (when a perpendicular height is available); (3) the total area of the polygon equals the sum of the areas of all component triangles; (4) this decomposition strategy is the mathematical foundation used by land surveyors when measuring irregularly shaped plots.
Skills	Students are able to: (1) decompose any irregular quadrilateral or pentagon into triangles by drawing non-crossing diagonals; (2) identify which triangle area formula applies given the available measurements; (3) calculate the area of each component triangle accurately; (4) sum component areas to find the total area of the irregular polygon; (5) present and justify their decomposition strategy to peers.
Attitudes	Students appreciate that mathematics is a practical tool for solving real land-ownership and inheritance disputes in Kenya, and value precision and systematic thinking when multiple formulae are available.
Key Inquiry Question	How do we apply area of polygons in real-life situations?
Purpose in Storyline	This is Lesson 7 of 8 in the evidence-gathering sequence. Students have previously explored the sine formula (½ab sin C), Heron's formula, and areas of parallelograms, trapeziums, kites, and regular polygons. In this lesson, learners return fully to the anchoring phenomenon — the surveyor's five-sided shamba in Nakuru — and apply all accumulated tools to decompose and solve the irregular polygon problem. They also solve the agricultural officer's quadrilateral farm plot and the Eldoret school's pentagonal mural wall, consolidating all prior learning into one unifying strategy. By the end of this lesson, students can answer the central driving question for any irregular polygon, preparing them for the final lesson's summative application.
Safety Notes	No physical safety concerns. Remind students to use rulers and protractors carefully when drawing decomposition diagrams. Ensure all drawn diagonals are labelled clearly to avoid computational errors during area summation.

B. LESSON OVERVIEW

Students open by revisiting the Nakuru surveyor's five-sided shamba plot — the anchoring phenomenon — and confront the key question: 'We now have all these formulae, but how do we actually solve an irregular five-sided plot?' Through a guided Predict-Observe-Explain cycle, learners discover that decomposing a polygon into triangles by drawing diagonals is the master strategy that unlocks the problem. Working in groups, they decompose two new irregular polygons (an agricultural officer's quadrilateral plot near Eldoret and a pentagonal mural wall at a school in Nakuru), choose the correct formula for each triangle, calculate component areas, and sum them. A class Model-Building session then updates the shared anchor diagram on the Driving Question Board, showing the complete solution pathway for the surveyor's shamba. The lesson integrates NGSS SEP 2 (Developing and Using Models), SEP 4 (Analysing and Interpreting Data), and SEP 6 (Constructing Explanations).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy

Predict Phase 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Polygon Classification in the Coordinate Plane (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students are presented with a scale diagram of the surveyor's five-sided shamba from the anchoring phenomenon (pentagon ABCDE with labelled side lengths and some included angles). They individually write answers to two prompts in their exercise books: (1) 'Which formula or formulae from our previous lessons could help the surveyor find the area of this whole plot? Explain your choice.' (2) 'Draw one or more lines on the diagram to show how you would break this shape into smaller, solvable parts.' Students then share predictions with a seat partner before the teacher takes a cold-call sample.	Display the pentagon ABCDE diagram on the board (or printed worksheet) with the following measurements: AB = 42 m, BC = 38 m, CD = 45 m, DE = 30 m, EA = 50 m; angle ABC = 78°, angle BCD = 65°. Say verbatim: 'You have spent six lessons building a toolkit of area formulae. Today, the surveyor is standing at the Nakuru shamba and needs to use that entire toolkit at once. Before we do anything, I want you to predict — individually and in silence — which tools you will reach for and how you will break this shape apart.' Allow 4 minutes of silent individual thinking. Then say: 'Turn to your partner. Compare your diagrams. Do you agree on where to draw the dividing lines?' Allow 2 minutes of pair discussion. Cold-call 3 students (spread across confidence levels): 'Amina, show us where you drew your lines and tell us why.' After each response, allow 10 seconds of wait time before commenting. Record the three different decomposition strategies on the board without evaluating them yet. Say: 'Hold on to these ideas — we are about to find out which strategy works best and why.'	Strategy: Activate-and-Surface Prior Knowledge. By asking students to predict decomposition before instruction, the teacher surfaces existing mental models and misconceptions (e.g., students who try to apply a single polygon formula directly). This creates productive cognitive dissonance that the Observe phase will resolve, making new learning more memorable.	Observable indicator: Each student has a labelled diagram with at least one diagonal drawn, and at least one formula name written in their book. Teacher scans books during cold-call phase. Look for: (1) students who draw diagonals correctly from one vertex (correct); (2) students who draw crossing diagonals that create overlapping triangles (misconception to address in Explain phase); (3) students who write 'use Heron's formula for the whole shape' (misconception — Heron's applies to triangles only, not pentagons directly).
Observe Phase 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:**	Students work in groups of four to complete a structured 'Decomposition Discovery' activity using three worked examples presented on printed cards (or the board). Card 1: A quadrilateral PQRS (agricultural officer's plot near Eldoret) with PQ = 60 m, QR = 45 m, PR = 75 m (diagonal), RS = 50 m, SP = 55 m, PS = 70 m (diagonal). Card 2: A step-by-step demonstration (teacher-led) of how pentagon ABCDE from the	Distribute the three cards and say verbatim: 'Your task right now is NOT to calculate. Your task is to observe, label, and decide. For each triangle in each decomposition, write the name of the formula you will use and circle the measurements you will plug in.' Circulate to each group. At each group, ask: 'Why did you choose that formula for this particular triangle — what measurements do you have?' Allow 10–12 seconds	Strategy: Structured Observation with Evidence Recording. By separating the observation step (identify and label) from the calculation step, this strategy forces students to engage in NGSS SEP 4 (Analysing and Interpreting Data) before computing. Students build the habit of reading the 'data landscape' of a problem — understanding what is given and what formula that unlocks — rather	Observable indicator: Each group's annotated diagram shows (a) correct non-overlapping triangles, (b) correct formula labels matched to available measurements. Teacher checks specifically: Does each group correctly identify that triangle ACD in the pentagon example has three known sides (suitable for Heron's formula) while triangle ABC has two sides and an included angle (suitable for ½ab sin C)? Groups that label both as

Polygon Classification in the Coordinate Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	phenomenon is split into three triangles by drawing diagonals AC and AD from vertex A, creating triangles ABC, ACD, and ADE. Card 3: A partially solved example of a pentagonal mural wall at a Nakuru school (side lengths only — no angles) requiring Heron's formula for each triangle. Groups observe and record: (a) how many triangles result from each decomposition; (b) which measurements are available for each triangle; (c) which formula is appropriate for each triangle and why. Groups annotate their diagrams with formula labels before any calculation begins.	of wait time after each question before accepting an answer. After 10 minutes, pause the class and conduct a whole-class 'gallery check': cold-call 4 different groups to report one triangle each, building a master table on the board with columns: Triangle | Known Measurements | Formula to Use. Say: 'Notice that in one polygon we might need two different formulae — Heron's for one triangle, the sine formula for another. The surveyor in Nakuru faces exactly this situation. This is why we built a toolkit, not just one formula.' Highlight the master table and leave it visible for the Explain phase.	than pattern-matching to a memorised procedure.	'Heron's formula' need targeted questioning: 'Do you know angle B? Then which formula allows you to use two sides and an angle?'
Explain Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Polygon Classification in the Coordinate Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students now carry out the full calculations for two problems: (A) The agricultural officer's quadrilateral PQRS plot near Eldoret — decomposed into two triangles, with one triangle solved using ½ab sin C and the other using Heron's formula; (B) The pentagonal mural wall at the Nakuru school — decomposed into three triangles, all solved using Heron's formula. Each group records full working in their books, including: decomposition diagram, formula stated, substitution shown, area of each triangle, and total polygon area. Groups then rotate to check another group's working (peer-marking with a provided answer key) before a teacher-led whole-class consolidation connects both solutions back to the anchoring phenomenon.	Say verbatim: 'Now we calculate. Work through Problem A first — quadrilateral PQRS. Write the formula, show every substitution, and box your final answer in square metres. You have 12 minutes.' Circulate constantly. At each group, ask at least one targeted question: 'You found s for triangle PQR — what does s represent physically in the context of this farm plot?' and 'When you calculated ½ × PQ × QR × sin(PQR), what would happen to the area if angle PQR were 90°? Would it be larger or smaller than now?' (Wait 12 seconds after each question.) After 12 minutes, cold-call one group to present Problem A on the board. Say: 'Class, as this group presents, track their working against yours. If you see a difference, put your hand up.' Address errors immediately. Then say: 'Now Problem B — the pentagonal mural wall. Three triangles, all Heron's formula. You have 15 minutes.' After	Strategy: Worked Example + Peer Critique (NGSS SEP 6 — Constructing Explanations). Students construct a full explanation of the solution pathway, then immediately subject it to peer critique. This dual process — explaining and critiquing — forces students to articulate their reasoning precisely, exposing gaps in understanding that silent individual work would not surface. Connecting the final calculation back to the anchoring phenomenon (the surveyor's shamba) gives the mathematics a narrative anchor, improving retention.	Observable indicator: Student books show (1) a correctly labelled decomposition diagram, (2) the correct formula stated before substitution, (3) correct calculation of s (for Heron's) or correct identification of the included angle (for sine formula), (4) correct summation of triangle areas. Teacher specifically checks: Does the student round intermediate values consistently? Is the final answer expressed in square metres (m^2)? Any student whose total polygon area for PQRS differs by more than 5% from the answer key should be asked to re-check their value of s or their sine calculation before the lesson ends.

		calculations, conduct whole-class consolidation. Draw the original pentagon ABCDE (surveyor's shamba) on the board and say verbatim: 'This is where we started seven lessons ago. The surveyor was stuck. Now, using exactly what you just did for PQRS and the mural wall, I want one volunteer to walk up and show the class how to solve the actual shamba problem.' Cold-call a student who showed strong working. Ask the class: 'Do you agree with each step? Thumbs up, thumbs sideways, or thumbs down.' Address any thumbs-sideways or thumbs-down responses with a second cold-call.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Polygon Classification in the Coordinate Plane (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each group receives two sticky notes. On the first (green), they write one thing they can now definitively answer about the driving question: 'How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot?' On the second (orange), they write one remaining question or uncertainty. Groups post both notes on the class Driving Question Board. The teacher facilitates a 5-minute whole-class read-through, grouping green notes by theme (decomposition strategy, formula selection, accuracy of measurement) and highlighting any orange notes that will be addressed in Lesson 8.	Hand out sticky notes and say verbatim: 'Six lessons ago, when we first saw the surveyor standing at the shamba, you had questions you could not yet answer. Today, what CAN you now tell that surveyor? Write it on green. What do you still wonder — about accuracy, about real surveying tools, about larger plots? Write that on orange.' Allow 4 minutes of quiet writing. Then say: 'Come up and post your notes. Let's read the green ones aloud together.' Read 5–6 green notes aloud, grouping them on the board. Say: 'Look at these — you have built a complete solution pathway: decompose, identify your measurements, choose your formula, calculate each triangle, sum the areas. That IS how a real surveyor works.' Then point to the orange notes: 'These remaining questions — about GPS measurements, about what happens when you don't even know all the sides — that is exactly where Lesson 8 will take	Strategy: Driving Question Board Synthesis (NGSS SEP 7 — Engaging in Argument from Evidence). The DQB update makes the progression of understanding visible and public. Students see — literally, on the wall — how their collective knowledge has grown across seven lessons. This metacognitive moment reinforces the storyline of the unit and creates anticipation for Lesson 8's final resolution.	Observable indicator: Green sticky notes contain specific, accurate mathematical statements (e.g., 'We can decompose the five-sided shamba into three triangles and use Heron's formula or the sine formula for each'). Vague statements (e.g., 'We know how to find area now') indicate surface-level understanding — the teacher should prompt: 'Can you name the specific strategy you would use?' Orange notes reveal genuine curiosity or conceptual gaps that inform Lesson 8 planning.

		us.'		
Model Building Phase 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Polygon Classification in the Coordinate Plane (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Students retrieve their cumulative 'Surveyor's Toolkit' model — a structured diagram they have been building across all seven lessons showing the decision tree: 'What measurements do I have? → Which formula do I use?' They now add a final branch to the decision tree: 'If the shape is an irregular polygon → Draw diagonals to decompose into triangles → Apply the toolkit to each triangle → Sum all triangle areas = Total polygon area.' Students also annotate the original shamba diagram (pentagon ABCDE) with their full solution: the three triangles, the formula used for each, the area of each triangle, and the grand total area in m^2. This annotated diagram is their capstone model for the unit.	Say verbatim: 'Open your Surveyor's Toolkit model from Lesson 1. Every lesson, we have added one new branch. Today we add the final branch — the one that ties everything together.' Display a blank version of the decision tree on the board with the new branch empty. Say: 'Individually, complete the new branch in your own model. Then annotate the ABCDE shamba diagram with the full solution. You have 8 minutes.' After 8 minutes, cold-call two students to share their completed decision tree branch. Ask the class: 'Is this branch complete? Could a land surveyor in Nakuru pick up this model and use it to solve any irregular plot — not just five-sided ones?' Wait 12 seconds. Say: 'What if the plot had eight sides? Would the strategy change?' (Expected answer: No — decompose into more triangles, same process.) Conclude: 'This is the power of decomposition. One strategy, infinite shapes. That is mathematical thinking.'	Strategy: Cumulative Model Revision (NGSS SEP 2 — Developing and Using Models). The Surveyor's Toolkit decision tree is a living model that students have refined across all seven lessons. Each revision makes prior learning visible and shows how new knowledge extends rather than replaces earlier understanding. Adding the final 'irregular polygon' branch completes the model and gives students a portable, self-authored reference tool they can use in assessments and in real life.	Observable indicator: The student's annotated ABCDE shamba diagram shows (1) correct diagonal placement creating three non-overlapping triangles, (2) each triangle labelled with the formula used, (3) correct intermediate calculations, (4) a correct grand total in m^2. The decision tree branch correctly states the decomposition strategy as the entry point for any irregular polygon. Students who cannot complete the annotation independently should be paired with a peer explainer before Lesson 8.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the Predict phase, how many students instinctively drew diagonals from a single vertex to create non-overlapping triangles? What does this tell you about the readiness of the class to handle the final Lesson 8 summative task? (2) In the Explain phase, did students correctly identify which formula to apply based on available measurements, or did most default to Heron's formula for every triangle regardless of given information? If the latter, plan a brief 5-minute recap at the start of Lesson 8 contrasting the two formulae with a side-by-side example. (3) Review the orange sticky notes from the DQB update. Do they reveal conceptual gaps about polygon decomposition itself (e.g., 'Can I draw diagonals from any vertex?') or practical concerns about measurement accuracy in the field? The former must be addressed in Lesson 8; the latter can be acknowledged as an authentic extension into surveying technology. (4) Were all groups able to correctly sum the triangle areas to reach a consistent total for the PQRS quadrilateral? If more than two groups had significant errors, consider starting Lesson 8 with a brief peer-teaching moment where accurate groups explain their working to groups that struggled. (5) Did the cumulative Surveyor's Toolkit model feel meaningful to students, or was it treated as a mechanical add-on? If students engaged with it superficially, use the first five minutes of Lesson 8 to have students verbally 'walk a partner through' the entire decision tree from start to finish, using the surveyor's shamba as the navigating example.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that the surveyor's five-sided shamba in Nakuru — and both the agricultural officer's quadrilateral plot near Eldoret and the pentagonal mural wall at the Nakuru school — could not be solved as a single shape using one formula. When we drew diagonals from one vertex, the irregular polygons broke apart into familiar triangles. We observed that different triangles within the same polygon sometimes required different formulae depending on which measurements were available.
What did I learn?	We learned that any irregular polygon can be decomposed into non-overlapping triangles by drawing diagonals from a single vertex. The number of triangles formed is always two fewer than the number of sides ($n - 2$). We learned to select the correct area formula for each triangle based on available data: ½ab sin C when two sides and the included angle are known, Heron's formula ($A = \sqrt{[s(s-a)(s-b)(s-c)]}$) when all three sides are known, and ½ × base × height when a perpendicular height is available. The total area of the irregular polygon equals the sum of all triangle areas.
How does this explain the phenomenon?	We explained that the land surveyor in Nakuru can calculate the exact area of the five-sided shamba by drawing two diagonals (AC and AD) from vertex A, decomposing the pentagon into triangles ABC, ACD, and ADE. For each triangle, the surveyor applies whichever formula matches the available field measurements. Summing the three triangle areas gives the total plot area in square metres. This decomposition strategy works for any irregular polygon, regardless of the number of sides, making it the universal tool in the surveyor's mathematical toolkit.

LESSON 8 (80 minutes): Consolidation, Model Revision & Real-World Applications: Answering the Driving Question	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all polygon area strategies learned across the unit, revise the running model, and construct a final, complete explanation for how a land surveyor in Nakuru can calculate the exact area of any irregularly shaped farm plot regardless of the measurements available.
Knowledge	Learners demonstrate integrated knowledge of: (a) the sine rule for triangle area ($A = ½ab \sin C$), (b) Heron's formula ($A = \sqrt{[s(s-a)(s-b)(s-c)]}$), (c) area formulas for parallelograms, trapezoids, kites, and regular polygons ($A = ns^2/[4 \tan(180°/n)]$), and (d) decomposition of irregular polygons into triangles and quadrilaterals.
Skills	Learners select and apply the most appropriate area formula for a given polygon, decompose irregular polygons into manageable sub-shapes, sum component areas accurately, and justify formula choice using available measurements — mirroring the professional decisions made by a land surveyor.
Attitudes	Learners appreciate the power of mathematics as a practical problem-solving tool in real Kenyan land management contexts, demonstrate persistence when tackling multi-step composite area problems, and value collaborative peer review in refining mathematical reasoning.
Key Inquiry Question	How do we apply area of polygons in real-life situations?
Purpose in Storyline	This is the final lesson of the unit. Learners have spent seven lessons gathering mathematical tools — from ½bh to Heron's formula to regular polygon decomposition. Today they bring every tool to bear on the original Nakuru shamba problem, produce a final written explanation for the driving question, compare their Lesson 1 naive model against their fully revised model, and close the Driving Question Board. The lesson celebrates how the full toolkit they have built is exactly what the surveyor needs.
Safety Notes	No physical hazards. Ensure group presentations are conducted respectfully; remind learners of school values of dignity and mutual respect when giving peer feedback.

B. LESSON OVERVIEW
In this 80-minute consolidation lesson, learners revisit the anchoring phenomenon — a land surveyor in Nakuru County tasked with calculating the area of an irregularly shaped five-sided shamba — and construct their final, complete explanation for the unit driving question. Working in groups, learners apply the full range of polygon area strategies (sine formula, Heron's formula, quadrilateral formulas, regular polygon formula, and irregular polygon decomposition) to a rich mixed-context problem set drawn from Kenyan land, architecture, and design scenarios. Groups then present their revised polygon area models, comparing Lesson 1 sketches against final models to make learning growth visible. The lesson closes with a whole-class Driving Question Board (DQB) review ceremony that traces every sticky note from initial wonder to final explanation, and a teacher-led metacognitive reflection on the mathematical journey from ½bh to composite decomposition strategies. Formative assessment is embedded throughout via group whiteboards, gallery walk peer feedback, and a final individual exit explanation card.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Learners receive a single A4 'Surveyor's Toolkit Card' showing a	Distribute Surveyor's Toolkit Cards face-down. Say: 'Before we solve	Knowledge Inventory / Retrieval Practice — by asking learners to	Teacher circulates during individual recall and notes which

📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Polygon Classification in the Coordinate Plane (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	blank table with five rows labelled: Triangle (no height given), Triangle (all three sides given), Quadrilateral (parallelogram/trapezium/kite), Regular Polygon, Irregular Polygon. Working individually for 3 minutes, learners fill in the formula they would use for each case and the minimum measurements required — drawing entirely from memory. They then share with a shoulder partner for 2 minutes and reconcile any differences. The class is cold-called for responses before the teacher reveals a colour-coded answer key on the board. This re-activates the entire unit's knowledge base and surfaces any lingering gaps before the final explanation task.	the big problem today, I want you to prove to yourself how far you have come. Flip your card. Fill in every row from memory — no notes. You have 3 minutes. Go.' Set a visible timer. After 3 minutes say: 'Compare with your partner. Where do you agree? Where do you disagree? 2 minutes.' After partner share, cold-call 5 different students — one per row — using name sticks. For each response, ask the class: 'Hands up if you had something different for this row.' Address discrepancies immediately. Display the colour-coded answer key and say: 'These are your five professional tools. Today you use all of them to solve the Nakuru shamba problem once and for all.'	recall formulas from memory before instruction, this strategy strengthens long-term retention (the testing effect) and makes prior knowledge explicit, so learners can consciously select tools during the final explanation task.	rows are left blank or incorrectly filled. Observable indicator: learner can write $A = ½ab \sin C$ and $A = \sqrt{[s(s-a)(s-b)(s-c)]}$ and $A = ns^2/[4 \tan(180°/n)]$ without prompting. Any learner with more than two blank rows is paired with a stronger peer for the group task.
Observe Phase 📹 **VIDEO:** [Overview of Ellipses, Trapezoids, Rhombi, and Polygons] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Polygon Classification in the Coordinate Plane (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The class stands and gathers around the Driving Question Board (DQB) that has been built across all eight lessons. The teacher reads aloud the driving question: 'How can a land surveyor in Nakuru calculate the exact area of any irregularly shaped farm plot, no matter what measurements are available?' Learners are given 2 minutes to silently read every sticky note on the board — questions asked in earlier lessons, partial answers posted along the way, and vocabulary terms added over time. Each learner then physically moves one sticky note from the 'Still Wondering' column to the 'We Can Now Explain This' column if they believe the unit has answered it, attaching a brief written justification on the back. The teacher facilitates a 4-minute discussion about the migration of notes, highlighting how the unit storyline moved from ½bh → sine	Stand beside the DQB and say: 'This board has been our scientific and mathematical memory for the whole unit. Every question, every breakthrough, every tool — it is all here.' Give 2 minutes of silent reading time (enforce silence with a visible timer). Then say: 'If you believe the unit has given you enough tools to answer a question on this board, walk up, take the sticky note, write your justification on the back, and move it to the green column. You have 2 minutes.' After migration, ask 3 learners (cold-call by name): 'Which note did you move, and in one sentence, what evidence allowed you to move it?' Conclude by pointing to any notes still in 'Still Wondering' and saying: 'Let's see if today's final task closes those gaps too.'	Driving Question Board Evidence Audit — this strategy makes the cumulative knowledge journey visible and tangible. By physically moving sticky notes, learners perform a metacognitive audit of their own understanding, connecting each prior lesson's evidence to the overarching driving question.	Teacher observes which sticky notes are moved and reads the justifications written on the back. Observable indicator: learner's written justification references a specific formula or strategy learned in a named lesson (e.g., 'In Lesson 5 we learned Heron's formula, which means we do not need a height at all — only the three side lengths'). Notes that are not moved flag conceptual gaps for targeted support during Phase 3.

	formula → Heron's formula → quadrilateral formulas → regular polygons → composite decomposition.			
Explain Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Polygon Classification in the Coordinate Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of four receive the 'Nakuru County Land & Design Challenge' problem card — a rich, mixed-context problem set containing four interconnected scenarios all tied to the anchoring phenomenon: (1) The five-sided shamba: given three side lengths and two included angles, decompose the plot into three triangles, apply the sine formula and Heron's formula as appropriate, and find the total area in hectares. (2) A parallelogram-shaped maize field in Kericho: find its area given base and height, then find the cost of ploughing at KSh 3,500 per hectare. (3) A decorative kite-shaped tile design for a new community centre in Kisumu: find the tile's area given both diagonals. (4) A regular hexagonal water harvesting tank base in Nakuru: find its floor area using $A = ns^2/[4 \tan(180°/n)]$. Groups work collaboratively for 18 minutes on whiteboards or large paper, showing full working for each sub-problem. Each group then presents one assigned scenario (2 minutes per group) and the class provides structured peer feedback using the 'WWW/EBI' (What Went Well / Even Better If) protocol. The teacher facilitates connections between each scenario and the surveyor's toolkit.	Distribute problem cards and say: 'These four problems all live in the same world as the Nakuru surveyor. Your group must solve all four, but each group will present one. I will assign presentations randomly at the end — so everyone must understand every problem.' Assign scenarios to groups after 18 minutes using random selection (name sticks in a cup). During group work, circulate every 3 minutes. Use targeted prompts: For groups stuck on Problem 1: 'Which diagonal do you draw first? What does that diagonal split the polygon into?' For groups stuck on Problem 4: 'What is n here? What is s? Substitute carefully.' After each 2-minute presentation, say: 'Class, give this group one WWW and one EBI — raise your hand when ready.' Cold-call 2 students for feedback after each presentation. After all four presentations, say: 'Notice what we just did — we just solved the exact problem the Nakuru surveyor was facing. Every formula you used exists because a real person, somewhere in Kenya, needed it.'	Jigsaw Presentation with Peer Feedback — by requiring all groups to solve all problems but each to present only one, this strategy creates authentic interdependence and whole-class accountability. The WWW/EBI protocol structures peer feedback to be specific and constructive, reinforcing mathematical communication skills.	Teacher checks group whiteboards at the 10-minute mark. Observable indicators: (a) Problem 1 solution shows correct polygon decomposition into triangles with labelled sub-triangles; (b) correct substitution into $A = ½ab \sin C$ or Heron's formula with units converted to hectares; (c) Problem 4 shows correct substitution of $n = 6$ and the given side length into the regular polygon formula. A group that cannot decompose Problem 1 correctly receives an immediate teacher scaffold: a pre-drawn diagonal to prompt the first decomposition step.
Driving Question Board (DQB) Creation 📹 **VIDEO:**	The class reconvenes at the DQB for the final ceremony. The teacher reads the driving question aloud one last time, and learners are invited to co-construct a final class	Stand at the DQB and say: 'We have arrived. Today we close this board — not because learning is over, but because we have answered the question it was	Model Revision with L1 vs. L8 Comparison — by returning to the original Lesson 1 model and explicitly comparing it against the final model, this strategy makes	Teacher collects or photographs final running models. Observable indicator: the Lesson 8 panel contains a clearly labelled decision tree covering all five formula

Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Polygon Classification in the Coordinate Plane (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	answer on a large sheet of paper that will be permanently posted beside the DQB. Working as a whole class, learners contribute sentences verbally while the teacher scribes. The final answer must include: (a) the decision-tree logic for formula selection (if height is known → ½bh; if two sides and included angle known → ½ab sin C; if three sides known → Heron's; if regular polygon → ns²/[4 tan(180°/n)]; if irregular → decompose); (b) a reference to the Nakuru shamba as the specific real-world context; and (c) at least one numerical example from the Phase 3 problem set. Each learner then updates their own running polygon area model (started in Lesson 1) by adding a final 'Lesson 8' panel showing the complete decision tree and the five-sided shamba solution. Learners compare their Lesson 1 model (which likely showed only ½bh) against their final model, and write one sentence at the bottom: 'Between Lesson 1 and Lesson 8, my model grew because...'	asking. Let's write the answer together.' Point to the driving question banner and ask: 'If the Nakuru surveyor asked you right now — what would you tell them? Who wants to start?' Take contributions from at least 6 learners (cold-call 3, accept 3 volunteers). Scribe the class answer on a large poster, shaping contributions into a coherent decision-tree paragraph. Then say: 'Now open your running model. Add a Lesson 8 panel. Show the decision tree and the shamba solution. And at the bottom, write one sentence about how your model grew.' Allow 6 minutes. Ask 3 learners (cold-call) to read their 'my model grew because...' sentence aloud. Affirm growth publicly: 'You walked in here in Lesson 1 knowing one formula. You walk out knowing five, and you know when to use each one. That is what it means to think like a mathematician — and like a surveyor in Nakuru.'	conceptual growth visible and consolidates the unit's learning trajectory into a single artefact. The co-constructed class answer on the poster creates shared ownership of the final explanation.	pathways, and the 'my model grew because...' sentence names specific formulas or strategies learned during the unit. The sentence 'my model grew because I now know Heron's formula and can find area without a perpendicular height' is the target level of response.
Model Building Phase 📹 **VIDEO:** Overview of Ellipses, Trapezoids, Rhombi, and Polygons Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Polygon Classification in the Coordinate Plane (Flexbook)	Each learner individually completes a structured 'Surveyor's Final Report Card' — a one-page exit task that functions as both the final formative assessment and the culminating model artefact. The card has three sections: (1) GIVEN A PROBLEM — a short, unseen irregular quadrilateral (a trapezium-shaped plot in Eldoret, given parallel sides and height) to solve independently with full working shown; (2) EXPLAIN YOUR CHOICE — one paragraph explaining why they chose the trapezium formula and not Heron's formula for this problem; (3)	Distribute the Surveyor's Final Report Card and say: 'This is your final individual task. Section 1 is a new problem — unseen. Section 2 asks you to justify your formula choice. Section 3 asks you to reflect on the whole journey. You have 12 minutes. Work in silence.' Set a visible 12-minute timer. Circulate silently — do not prompt on Section 1; observe and note who is struggling with the trapezium area calculation. At the 12-minute mark, say: 'Pens down on Sections 1 and 2. Now find a partner and share your Section 3 reflections. You have 3 minutes.'	Structured Exit Explanation with Justification — by requiring learners to both solve a problem and explain their formula choice in writing, this strategy assesses not just procedural fluency but conceptual understanding and transfer. The three-part reflection section ensures metacognitive closure, connecting the mathematical tools to the real-world phenomenon that anchored the unit.	Teacher reviews collected Surveyor's Final Report Cards after the lesson. Observable indicators for mastery: (a) correct trapezium area calculation with units; (b) explanation paragraph explicitly states why Heron's formula is inappropriate here (it requires three sides of a triangle, not a quadrilateral) and correctly identifies the trapezium formula; (c) Section 3 reflection demonstrates understanding of the connection between polygon area mathematics and real-world land surveying in Kenya. Cards that show procedural accuracy but

Source: CK-12 🔍 Search ARES for similar readings	REFLECT ON THE JOURNEY — three sentence stems to complete: 'Before this unit, I thought area was...', 'The tool I found most powerful was... because...', 'A land surveyor in Nakuru would need this mathematics because...'. Learners work in silence for 12 minutes. The final 3 minutes are used for a think-pair-share on the reflection sentences only.	After 3 minutes, cold-call 2 learners to read their 'A land surveyor in Nakuru would need this mathematics because...' sentence to the class. Collect all cards as the summative unit record. Close with: 'The shamba in Nakuru has been measured. The siblings can now share their inheritance fairly — because you know the mathematics that makes that possible. Well done.'		weak justification (Section 2) flag learners who need further work on mathematical communication in the next unit.

D. TEACHER REFLECTION

After this final lesson, reflect on the following questions: (1) Could every learner articulate — in writing — a clear decision tree for formula selection, or did some learners still rely on memorised steps without conceptual understanding of why each formula applies? (2) When comparing Lesson 1 and Lesson 8 models, which conceptual leap proved hardest for most learners — moving from ½bh to the sine formula, or from triangle decomposition to irregular polygon decomposition? What does this tell you about where to invest instructional time in future cycles of this unit? (3) Did the real-world Nakuru shamba context sustain learner engagement across all eight lessons, or did it feel forced in some lessons? Which Kenyan context (land surveying, tile design, water tank, maize field) generated the richest mathematical discussion? (4) Review the Surveyor's Final Report Cards: what percentage of learners successfully justified their formula choice (Section 2) versus simply solving the problem (Section 1)? This gap is a direct measure of the depth of conceptual understanding developed across the unit. (5) Did the Driving Question Board ceremony feel meaningful to learners, or procedural? Consider how to involve learners more actively in curating the DQB throughout the unit — not just at the close — in your next delivery of this sub-strand.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed (across the unit) that the Nakuru surveyor could not apply the standard ½ × base × height formula to the irregularly shaped five-sided shamba because no perpendicular height was directly measurable from the given field data. They also observed that the plot could be physically decomposed into triangles and quadrilaterals by drawing internal diagonals, making the problem tractable.
What did I learn?	Students learned that any polygon area problem can be solved by selecting from a toolkit of five strategies: (1) ½bh when a perpendicular height is available; (2) $A = \frac{1}{2}ab \sin C$ when two sides and an included angle are known; (3) Heron's formula $A = \sqrt{s(s-a)(s-b)(s-c)}$ when all three sides of a triangle are known but no angle or height is given; (4) specific quadrilateral formulas for parallelograms, trapezoids, and kites; and (5) $A = ns^2/[4 \tan(180°/n)]$ for regular polygons. Irregular polygons are solved by decomposing them into these known shapes and summing the component areas.
How does this explain the phenomenon?	Students can now explain that a land surveyor in Nakuru can calculate the exact area of any irregularly shaped farm plot by (a) identifying which measurements are available at each boundary triangle or quadrilateral after decomposing the plot with internal diagonals, (b) selecting the appropriate formula for each component shape based on those available measurements, (c) calculating each component area and summing them, and (d) converting units to hectares for practical land use. The surveyor does not need a perpendicular height for every triangle — the sine formula and Heron's formula eliminate that requirement entirely, making the complete measurement of any polygon possible with the toolkit developed in this unit.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.